Signal Detection for User-Centric Network Massive MIMO System

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***Abstract*—In this paper, we investigate the signal detection for user-centric network (UCN) massive multi-input multi-output (mMIMO) system. We consider that the users are divided into multiple user groups (UGs). For each UG, leveraging the interference sparsity, we reveal that the performance of the minimum-mean-square-error (MMSE) detector can be guaranteed in the network mMIMO system by using the matched filtering (MF) outputs of the intra-group and interfering users. Then, with the base station (BS) connection sparsity, we reveal that the detection performance of each UG is primarily determined by a limited number of associated BSs. To facilitate practical application, we propose a straightforward user grouping method and outline the process for determining interfering users and associated BSs for each UG in UCN mMIMO systems. Then, we propose a user-centric detection method that decouples the detection process for each UG into two stages. In the first stage, local MF is performed at each associated BSs using local information. In the second stage, group-wise interference cancellation (IC) is carried out to obtain detection results at the primary serving BS (PSBS) of each UG, with information exchanged from auxiliary serving BSs (ASBSs). Simulation results confirm the effectiveness and computational efficiency of our proposed user-centric detection for the UCN mMIMO system.**

***Index Terms*—Base station connection sparsity, interference sparsity, signal detection, user-centric network massive MIMO.**

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I. Introduction

With the rapid development of communication applications, the network traffic and user access have witnessed explosive growth during the past ten years [1], which in turn drives the advancement of the wireless technologies [2]. Among the wireless technologies, massive multi-input multi-output (mMIMO) has evolved into one of the key technologies in the fifth generation (5G) networks [3], and is expected to maintain its crucial role in the sixth generation (6G) era [4]. By equipping a large number of antennas at the base station (BS), mMIMO can simultaneously serve a number of users in the same time-frequency resource [5], [6], [7], effectively enhancing the spectrum and energy efficiencies [8], [9].

For mMIMO systems, signal detection is of critical significance to harness the potential capacity gains of mMIMO, and has been widely investigated [10]. With large numbers of antennas and users, the maximum *a posteriori* (MAP), which is known as the optimal detection method, suffers from unaffordable computational complexity. Although approaches like sphere decoding [11] and local search [12] can achieve near-optimal performance with decreased complexity, their computational costs remain unacceptable in practice. To facilitate practical application, several methods have been proposed to reduce the complexity, such as the belief propagation (BP) [13], expectation propagation (EP) [14], approximate message passing (AMP) [15], and their variants [16], [17]. On the other hand, as the numbers of BS antennas and users increase, linear detectors such as zero-forcing (ZF) and minimum mean-squared error (MMSE) detectors have shown progressively significant advantages and garnered much attention in many scenarios due to their appealing low complexity [18], [19], [20].

The performances of the detection techniques discussed above have been validated for the most widely adopted cellular mMIMO systems [21], [22]. Moreover, the detection performance can be further improved by introducing advanced channel coding techniques. The research in [23] and [24] provides a comprehensive analysis of how Turbo coding can enhance detection performance, and [25] further deals with the probability of erasure computation. Nevertheless, cell-edge users in a cellular mMIMO system could suffer from poor quality of service (QoS) due to inherently weak channel conditions and strong inter-cell interference [26]. This issue would

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become more severe with the further increase in user numbers and the continuous reduction in cell size, necessitating further exploration into systems capable of delivering seamless coverage. To mitigate this limitation, network mMIMO system has been proposed in [27] to enhance the performance of cell-edge users via coherent joint processing [28], which has reemerged recently under the name of cell-free mMIMO system [29], [30], [31], [32]. To be specific, by exchanging the information between BSs via backhaul links, each user can be served by all the BSs in the network mMIMO system, thereby ensuring seamless service coverage [33]. As a result, joint detection in network mMIMO system enables more efficient interference management and improves signal quality at the cell edge, delivering enhanced QoS for users in the system. In fact, many detection methods have been further explored and developed in the context of network mMIMO systems [34], [35], [36], [37], [38], demonstrating outstanding detection performance.

However, as the network size expands and the number of users increases, joint detection of all users by all BSs could lead to unaffordable computational complexity and become impractical. More recently, the user-centric rule has been introduced [39], leading to the user-centric network (UCN) mMIMO system proposed in [40]. In the UCN mMIMO system, the channel gains between each user and most BSs are negligible, indicating that the channel matrix is a sparse matrix inherently. This implies that each user can be served by only a limited number of neighboring BSs while maintaining the QoS [41]. This not only eliminates the notion of cell-edge, thereby enhancing the performances of cell-edge users, but also facilitates practical implementation [42]. In addition, the interference between users in different directions or remote users is relatively low, indicating that the channel Gram matrix exhibit inherent sparsity. Therefore, the detection performance of each user is expected to be determined by a small subset of interfering users in the network. Therefore, this system inherently adopts a distributed architecture, and can effectively reduce the dimension of the observation vector and the complexity of matrix operations for detection, alleviating both computational complexity and implementation challenges. Additionally, given the truth that cellular mMIMO systems have been widely deployed, the UCN mMIMO system offers a smoother and more feasible transition, presenting advantages in enhancing service quality for more cell edge users over wider areas [43].

The UCN mMIMO system has shown significantly lower computational complexity and satisfactory sum-rate performance in DL precoding [40]. In this paper, we aim to establish a rigorous theoretical foundation for UCN mMIMO systems, demonstrating its efficiency and optimality under certain conditions, thereby paving the way for a practical user-centric detection method. We consider that the users are divided into multiple user groups (UGs), and investigate the uplink (UL) signal detection for UCN mMIMO systems by leveraging the inherent interference and BS connection sparsities of each UG. The main contributions are summarized as follows.

- Leveraging the inherent interference sparsity, we demonstrate that the detection performance of the MMSE detector can be guaranteed by utilizing the matched filtering (MF) outputs of the intra-group and a small number of interfering users in the network mMIMO system. To be specific, using the MMSE detector, we prove that considering more interfering users can improve the detection performance. On the other hand, we also show that omitting the interference caused by majority of users in the system has little impact on the detection performance.
- By virtue of the BS connection sparsity, we reveal that the detection performance of each UG is primarily determined by a subset of BSs. Specifically, we quantify the contribution of each BS to the MSE performance of each UG, offering guidance for selecting the associated BSs. Additionally, we demonstrate that the BSs with weak channel connectivity with each UG have little influence on the detection performance of each UG.
- To facilitate practical application, we present a simple user grouping approach and establish effective criteria for selecting interfering users and associated BSs for each UG. On this basis, we introduce a user-centric detection method composed of two stages for UCN mMIMO systems. In the first stage, local matched filtering (MF) is carried out at each associated BS for each UG, followed by interference cancellation (IC) at the primary serving BS (PSBS) in the second stage, with necessary information shared from the auxiliary serving BSs (ASBSs).
- The complexity analyses show that the proposed signal detection for UCN mMIMO systems has a much lower computational complexity than the network mMIMO system. Simulation results further demonstrate the effectiveness and superiority of the user-centric detection for UCN mMIMO system.

TABLE I
LIST OF ABBREVIATIONS

Abbreviations	Meanings
ASBS	Auxiliary serving base station
BBU	Baseband unit
BER	Bit error rate
BS	Base station
BC	Base station cluster
MF	Matched filtering
mMIMO	Massive multi-input multi-output
MMSE	Minimum-mean-square-error
NMSE	Normalized mean-squared error
IC	Interference cancellation
IG	Interference group
PSBS	Primary serving base station
PSU	Primary served user
SSU	Secondary served user
UCN	User-centric network
UG	User group

The rest of this paper is organized as follows. In Section II, we leverage the interference and BS connection sparsities in group based signal detection, demonstrating that the detection performance of each UG is determined by a subset of interfering users and a limited number of BSs. The UCN

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mMIMO system and the user-centric detection approach are discussed in Section III. Simulation results are presented in Section VI to validate the superiority of the signal detection in UCN mMIMO systems. Finally, the conclusion is drawn in Section V. We have compiled the abbreviations frequently used throughout the paper and presented them with their full forms in Table I.

*Notations:*Boldface lowercase and uppercase letters represent the column vectors and matrices, respectively. We write conjugate transpose of matrix $\mathbf{A}$ as $\mathbf{A}^H$ while $\mathrm{tr}(\mathbf{A})$ and $\det(\mathbf{A})$ denote the matrix trace and determinant of $\mathbf{A}$, respectively. $\Re\{\mathbf{A}\}$ means the real part of $\mathbf{A}$. Let $|a|$, $\|\mathbf{a}\|$ and $\|\mathbf{A}\|_\infty$ denote the magnitude of a, the norm of $\mathbf{a}$ and the maximum absolute value among all elements of $\mathbf{A}$, respectively. Let the mathematical expectation be $\mathbb{E}\{\cdot\}$. $\mathbf{I}_M$ denotes the $M \times M$ identity matrix, whose subscript may be omitted for brevity. $\mathbf{0}$ represents the vector or matrix whose elements are all zero. $\mathrm{diag}(\mathbf{a})$ represents the diagonal matrix with $\mathbf{a}$ along its main diagonal and $\mathrm{diag}(\mathbf{A})$ denotes the column vector of the main diagonal of $\mathbf{A}$. All notations written in calligraphic font, e.g.,$\mathcal{A}$, represent sets, and $\mathrm{card}(\mathcal{A})$ denotes the cardinality of set$\mathcal{A}$.

II. MMSE Detection With Interference Sparsity and BS Connection Sparsity

In this section, with the inherent interference and BS connection sparsities, we reveal that the performance of the MMSE detector is determined by a small subset of interfering users and a limited number of associated BSs in group based signal detection.

Consider a network mMIMO system with U single-antenna users served by B BSs, and each BS has N antennas. Notably, this system can be directly extended to multi-antenna user scenarios by leveraging the singular value decomposition (SVD) to decompose each user's channel into orthogonal eigenmodes, effectively transforming them into multiple virtual single-antenna users. Let $\mathcal{B} = \{1, 2, \cdots, B\}$ and $\mathcal{U} = \{1, 2, \cdots, U\}$ denote the sets of the BSs and users, respectively. Let $s_i \in \mathbb{C}$ denote the information-bearing signal transmitted by the i-th user with $\mathbb{E}\{s_i s_i^*\} = p_i$ and $\mathbb{E}\{s_i s_j^*\} = 0, j \neq i$, where p_i is the average transmit power of user i. $\mathbf{h}_{i,k} \in \mathbb{C}^{N\times 1}$ denotes the channel vector from the i-th user to the k-th BS.

A. Optimal MMSE Detection

We consider the MMSE detection for the network mMIMO system, where all the users are jointly served by all the BSs. It is assumed that the BSs are synchronized and linked via backhaul links in the network mMIMO system, which enables coherent joint detection.

Let $\mathbf{H}_k = [\mathbf{h}_{1,k}, \mathbf{h}_{2,k}, \cdots, \mathbf{h}_{U,k}] \in \mathbb{C}^{N\times U}$ and $\mathbf{s} = [s_1, s_2, \cdots, s_U]^T$. We assume that the perfect CSI is available at the BS side. The CSI can be acquired by the channel estimation and the channel prediction approach proposed in [7] and [44]. We assume perfect time and frequency synchronization and orthogonality between subcarriers. Then, we focus on the signal detection on a given subcarrier of an OFDM symbol, whose subscripts are omitted for brevity. The received signal $\mathbf{y}_k$ at BS k can be written as

$$\mathbf{y}_k = \sum_{j\in\mathcal{U}} \mathbf{h}_{j,k} s_j + \mathbf{z}_k = \mathbf{H}_k \mathbf{s} + \mathbf{z}_k, \tag{1}$$

where $\mathbf{z}_k$ is the independent and identically distributed (i.i.d.) circularly symmetric complex Gaussian noise vector distributed as $\mathcal{CN}(0, \sigma_z^2 \mathbf{I}_N)$. Then, the joint received signal can be expressed as

$$\underbrace{\begin{bmatrix}\mathbf{y}_1\\ \vdots \\ \mathbf{y}_B\end{bmatrix}}_{\mathbf{y}\in\mathbb{C}^{BN\times 1}} = \underbrace{\begin{bmatrix}\mathbf{H}_1\\ \vdots \\ \mathbf{H}_B\end{bmatrix}}_{\mathbf{H}\in\mathbb{C}^{BN\times U}} \mathbf{s} + \underbrace{\begin{bmatrix}\mathbf{z}_1\\ \vdots \\ \mathbf{z}_B\end{bmatrix}}_{\mathbf{z}\in\mathbb{C}^{BN\times 1}}, \tag{2}$$

which can be written in a more compact form as

$$\mathbf{y} = \mathbf{H}\mathbf{s} + \mathbf{z} \tag{3}$$

with $\mathbf{\Lambda} \triangleq \mathbb{E}\{\mathbf{s}\mathbf{s}^H\} = \mathrm{diag}(p_1, p_2 \cdots, p_U)$ and $\mathbf{z} \sim \mathcal{CN}(0, \sigma_z^2 \mathbf{I}_{BN})$. With the linear detector $\mathbf{V} \in \mathbb{C}^{BN\times U}$, the transmitted signal can be jointly recovered by

$$\hat{\mathbf{s}} = \mathbf{V}^H \mathbf{y} = \mathbf{V}^H \mathbf{H}\mathbf{s} + \mathbf{V}^H \mathbf{z}. \tag{4}$$

The conditional MSE is defined as

$$\begin{aligned}\mathrm{MSE}_{\mathrm{tot}} &= \mathbb{E}\left\{\left(\hat{\mathbf{s}} - \mathbf{\Lambda}^{-\frac{1}{2}}\mathbf{s}\right)^H \left(\hat{\mathbf{s}} - \mathbf{\Lambda}^{-\frac{1}{2}}\mathbf{s}\right) \mid \mathbf{H}\right\} \\ &= \mathrm{tr}\left(\mathbf{V}^H\left(\mathbf{H}\mathbf{\Lambda}\mathbf{H}^H + \sigma_z^2\mathbf{I}_{BN}\right)\mathbf{V} - 2\Re\left\{\mathbf{V}^H\mathbf{H}\mathbf{\Lambda}^{\frac{1}{2}}\right\} + \mathbf{I}\right),\end{aligned} \tag{5}$$

and the MSE performance of the i-th user with $\mathbf{v}_i$ is

$$\mathrm{MSE}(\mathbf{v}_i) = 1 + \mathbf{v}_i^H\left(\mathbf{H}\mathbf{\Lambda}\mathbf{H}^H + \sigma_z^2\mathbf{I}_{BN}\right)\mathbf{v}_i - 2\Re\left\{\sqrt{p_i}\mathbf{v}_i^H\mathbf{h}_i\right\}. \tag{6}$$

Then, the detector that minimizes the MSE (5) is given by

$$\begin{aligned}\mathbf{V}^H &= \mathbf{\Lambda}^{\frac{1}{2}}\mathbf{H}^H\left(\mathbf{H}\mathbf{\Lambda}\mathbf{H}^H + \sigma_z^2\mathbf{I}_{BN}\right)^{-1} \\ &= \left(\mathbf{\Lambda}^{\frac{1}{2}}\mathbf{H}^H\mathbf{H}\mathbf{\Lambda}^{\frac{1}{2}} + \sigma_z^2\mathbf{I}\right)^{-1}\mathbf{\Lambda}^{\frac{1}{2}}\mathbf{H}^H.\end{aligned} \tag{7}$$

Among all linear detectors, the MMSE detector is recognized as the optimal detector in the sense of not only minimizing the MSE but also maximizing the instantaneous SINR. Among all detection strategies, joint detection across all BSs is expected to be the optimal approach. Therefore, we refer to the jointly designed MMSE detector (7) as the optimal linear detector for this network mMIMO system.

By applying the optimal MMSE detector $\mathbf{V}^H$ to the received signal $\mathbf{y}$, the recovered signal (4) can be rewritten as

$$\hat{\mathbf{s}} = \left(\mathbf{\Lambda}^{\frac{1}{2}}\mathbf{H}^H\mathbf{H}\mathbf{\Lambda}^{\frac{1}{2}} + \sigma_z^2\mathbf{I}\right)^{-1}\mathbf{\Lambda}^{\frac{1}{2}}\mathbf{H}^H(\mathbf{H}\mathbf{s} + \mathbf{z}), \tag{8}$$

and the MSE is given by

$$\mathrm{MSE}_{\mathrm{tot}} = \mathrm{tr}\left(\mathbf{I}_U - \mathbf{\Lambda}^{\frac{1}{2}}\mathbf{H}^H\left(\mathbf{H}\mathbf{\Lambda}\mathbf{H}^H + \sigma_z^2\mathbf{I}_{BN}\right)^{-1}\mathbf{H}\mathbf{\Lambda}^{\frac{1}{2}}\right). \tag{9}$$

More specifically, the optimal detector $\mathbf{v}_i^\star$ of user i is the i-th column of $\mathbf{V}$, and the MSE of user i can be written as

$$\mathrm{MSE}(\mathbf{v}_i^\star) = 1 - p_i\mathbf{h}_i^H\left(\mathbf{H}\mathbf{\Lambda}\mathbf{H}^H + \sigma_z^2\mathbf{I}_{BN}\right)^{-1}\mathbf{h}_i. \tag{10}$$

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TABLE II
LIST OF COMMON NOTATIONS DURING DETECTION

Notation	Definition
$\mathbf{h}_{i,k} \in \mathbb{C}^{N\times 1}$	Channel vector from user i to BS k
$\mathbf{H}_k \in \mathbb{C}^{N\times U}$	Channel matrix from $\mathcal{U}$ to BS k
$\mathbf{h}_i \in \mathbb{C}^{BN\times 1}$	Channel vector from user i to $\mathcal{B}$
$\mathbf{H} \in \mathbb{C}^{BN\times U}$	Channel matrix from $\mathcal{U}$ to $\mathcal{B}$
$\mathbf{H}_{\mathcal{U}_x} \in \mathbb{C}^{BN\times U_x}$	Channel matrix from $\mathcal{U}_x$ to $\mathcal{B}$
$\mathbf{h}_{i,\mathcal{B}_y} \in \mathbb{C}^{B_yN\times 1}$	Channel vector from user i to $\mathcal{B}_y$
$\mathbf{H}_{\mathcal{U}_x,\mathcal{B}_y} \in \mathbb{C}^{B_yN\times U_x}$	Channel matrix from $\mathcal{U}_x$ to $\mathcal{B}_y$
$\bar{\mathbf{H}} \in \mathbb{C}^{U\times U}$	$\mathbf{H}^H\mathbf{H}$
$\bar{\mathbf{H}}_{\mathcal{U}_x} \in \mathbb{C}^{U_x\times U}$	$\mathbf{H}_{\mathcal{U}_x}^H\mathbf{H}$
$\mathbf{\Lambda}$	$\mathbb{E}\{\mathbf{s}\mathbf{s}^H\}$
$\mathbf{\Lambda}_{\mathcal{U}_x}$	$\mathbb{E}\left\{\mathbf{s}_{\mathcal{U}_x}\mathbf{s}_{\mathcal{U}_x}^H\right\}$
$\mathbf{s}$	The signal transmitted by all users
$\mathbf{s}_{\mathcal{U}_x}$	The signal transmitted by $\mathcal{U}_x$
$\mathbf{y}$	The received signal of all BSs
$\mathbf{y}_{\mathcal{B}_y}$	The received signal of BSs in $\mathcal{B}_y$
$\bar{\mathbf{y}}_{\mathcal{U}_x,\mathcal{B}_y} \in \mathbb{C}^{B_yN\times U_x}$	$\mathbf{H}_{\mathcal{U}_x,\mathcal{B}_y}^H\mathbf{y}_{\mathcal{B}_y}$
$\bar{\mathbf{y}}_{\mathcal{I}_m}$	$\mathbf{H}_{\mathcal{I}_m}^H\mathbf{y}$
$\tilde{\mathbf{y}}_{\mathcal{I}_m,\mathcal{B}_m}$	The reduced dimensional signal model with interference and BS connection sparsities
$\tilde{\mathbf{y}}_{\mathcal{I}_m}$	The reduced dimensional signal model with interference sparsity

Generally, we have $BN \geq U$ in the network mMIMO system. Therefore, the design and implementation complexities of this optimal MMSE detector are $\left(2BNU^2 + U^3\right)$ and BNU according to (7) and (4), respectively. Moreover, the CSI $\mathbf{h}_{i,k}, \forall i \in \mathcal{U}, k \in \mathcal{B}$, needs to be exchanged among all the BSs when designing (7). To be specific, for the fully connected network mMIMO system, the backhaul burden is BNU. As the size of the network expands and the number of antennas at each BS side increases, the overwhelming design and implementation complexities, and the considerable backhaul load may become unaffordable. Additionally, the inverse of large dimensional matrix in (7) is difficult to implement in practice and may suffer from numerical ill-conditioning.

B. MMSE Detection With Interference Sparsity

It is worth noting that the optimal MMSE detector $\mathbf{V}^H$ in (7) always does the MF first, and thus $\mathbf{V}^H \in \mathbb{C}^{U\times BN}$ actually lies in the column space of $\mathbf{H}^H \in \mathbb{C}^{U\times BN}$. In this way, when applying $\mathbf{V}^H$ to $\mathbf{y}$, we can define

$$\bar{\mathbf{y}} = \mathbf{H}^H\mathbf{y} = \bar{\mathbf{H}}\mathbf{s} + \bar{\mathbf{z}}, \tag{11a}$$

$$\bar{\mathbf{V}}^H = \left(\mathbf{\Lambda}^{\frac{1}{2}}\mathbf{H}^H\mathbf{H}\mathbf{\Lambda}^{\frac{1}{2}} + \sigma_z^2\mathbf{I}\right)^{-1}\mathbf{\Lambda}^{\frac{1}{2}}, \tag{11b}$$

where $\bar{\mathbf{y}}$ denotes the matched filtered signal and $\bar{\mathbf{V}} \in \mathbb{C}^{U\times U}$ is the MMSE detector using MF outputs for IC. Therefore, we refer to $\bar{\mathbf{V}}$ as the global MMSE-IC matrix for brevity. $\bar{\mathbf{H}} = \mathbf{H}^H\mathbf{H}$ and $\bar{\mathbf{z}} = \mathbf{H}^H\mathbf{z}$ are the channel Gram matrix and the matched filtered noise, respectively. Note that $\bar{\mathbf{H}}\mathbf{s}$ and $\bar{\mathbf{z}}$ are still independent with each other. Let

$$\mathbf{h}_i = \left[\mathbf{h}_{i,1}^T, \mathbf{h}_{i,2}^T, \cdots, \mathbf{h}_{i,B}^T\right]^T \in \mathbb{C}^{BN\times 1} \tag{12}$$

denote the stacked channel vector from user i to all the BSs. For clarity, Table II is presented to illustrate the definitions of commonly used notations during detection, where $\mathcal{U}_x$ and $\mathcal{B}_y$ represent arbitrary user and BS sets, respectively, with $\mathrm{card}\left(\mathcal{U}_x\right) = U_x$ and $\mathrm{card}\left(\mathcal{B}_y\right) = B_y$.

Then, the matched filtered signal (11a) can be written in a more detailed form as

$$\begin{bmatrix}\bar{y}_1\\ \bar{y}_2\\ \vdots\\ \bar{y}_U\end{bmatrix} = \begin{bmatrix}\mathbf{h}_1^H\mathbf{h}_1 & \cdots & \mathbf{h}_1^H\mathbf{h}_U\\ \mathbf{h}_2^H\mathbf{h}_1 & \cdots & \mathbf{h}_2^H\mathbf{h}_U\\ \vdots & \ddots & \vdots\\ \mathbf{h}_U^H\mathbf{h}_1 & \cdots & \mathbf{h}_U^H\mathbf{h}_U\end{bmatrix}\begin{bmatrix}s_1\\ s_2\\ \vdots\\ s_U\end{bmatrix} + \begin{bmatrix}\bar{z}_1\\ \bar{z}_2\\ \vdots\\ \bar{z}_U\end{bmatrix}. \tag{13}$$

Notice that the off-diagonal elements in the channel Gram matrix $\bar{\mathbf{H}}$ can reflect the strength of inter-user interference [45]. Specifically, when $\left|\left(\bar{\mathbf{H}}\right)_{i,j}\right| = \left|\mathbf{h}_i^H\mathbf{h}_j\right|$ is negligible, it can be seen as there is no inter-user interference between user i and j. For the mMIMO system with a large number of antennas at each BS, users in different directions from the BSs or located far apart inherently satisfy this condition. All these facts suggest that $\bar{\mathbf{H}}$ is numerically sparse, implying that the inter-user interference is limited for each user. We refer to this property as the *interference sparsity*. Then, we have the following proposition.

Proposition 1: Let the set of users be decomposed into M UGs

$$\mathcal{U} = \mathcal{U}_1 \cup \mathcal{U}_2 \cup \cdots \cup \mathcal{U}_M \tag{14}$$

with $\mathcal{U}_m = \{m_1, \cdots, m_{U_m}\}$ and $\mathrm{card}\left(\mathcal{U}_m\right) = U_m$. Then the MMSE detection using the MF outputs of intra-group users can maintain the MSE performance, if and only if

$$\mathbf{h}_i^H\mathbf{h}_j = 0, \forall i \in \mathcal{U}_m, \forall j \in \mathcal{U}_n, m \neq n. \tag{15}$$

Proof: The proof is available in Appendix A. □

Proposition 1 reveals that if the user channels within different UGs are mutually orthogonal, i.e., there is no interference between UGs, the performance of the MMSE detector can be guaranteed by using the MF outputs of the intra-group users for each UG. In particular, for scenarios where all user channels are mutually orthogonal, $\bar{\mathbf{H}}$ is a diagonal matrix and the MMSE detector (7) is reduced to the matched filter. Note that user grouping represents a more general system setup because the number of users in each group can be arbitrary. Moreover, Proposition 1 demonstrates that decomposing the users into multiple UGs for separate detection could be beneficial.

In practice, the number of users can be quite large, and in the worst case, there may be no users whose channels are perfectly orthogonal to those of other users. In this situation, user grouping can be achieved using any clustering method to ensure that users with strong interference are clustered together, while maintaining low interference between groups. Additionally, group based detection that only using the MF outputs of intra-group users loses the optimality in such scenarios, and more interfering users may need to be incorporated for each UG to improve the performance.

In the following, we focus on the signal detection for $\mathcal{U}_m$ with $i \in \mathcal{U}_m$. Let $\mathcal{I}_m = \mathcal{U}_m \cup \mathcal{U}_m^{\mathcal{I}}$ denote the set of users considered for detecting signals of $\mathcal{U}_m$ with $\mathrm{card}\left(\mathcal{I}_m\right) = I_m$, where $\mathcal{U}_m^{\mathcal{I}}$ is the set of additional interfering users. For clarity,

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TABLE III
LIST OF SET NOTATIONS

Notation	Definition
$\mathcal{B}$	The set of all BSs
$\mathcal{B}_m$	The BC of the m-th UG
$\bar{\mathcal{B}}_i$	The associated BSs of user i
$\tilde{\mathcal{B}}_m$	$\cup_{i\in\mathcal{U}_m}\bar{\mathcal{B}}_i$
$\mathcal{B}_k^{\mathrm{P}}$	The set of PSBSs associated with SSUs of BS k
$\mathcal{I}_m$	The IG of the m-th UG
$\mathcal{U}$	The set of all users
$\mathcal{U}_m$	The m-th UG
$\mathcal{U}_m^{\mathcal{I}}$	$\mathcal{I}_m\backslash\mathcal{U}_m$
$\bar{\mathcal{U}}_k$	The served users of the k-th BS
$\mathcal{U}_k^{\mathrm{P}}$	The PSUs of BS k
$\mathcal{U}_k^{\mathrm{S}}$	The SSUs of BS k

Table III is presented to illustrate the definitions of commonly used set notations throughout the paper.

Let

$$\bar{\mathbf{y}}_{\mathcal{I}_m}=\mathbf{H}_{\mathcal{I}_m}^H\mathbf{y}=\bar{\mathbf{H}}_{\mathcal{I}_m}\mathbf{s}+\mathbf{H}_{\mathcal{I}_m}^H\mathbf{z} \tag{16}$$

denote the partial MF outputs of (11) with $\mathbf{H}_{\mathcal{I}_m}$ and let $\bar{\mathbf{v}}_i(\mathcal{I}_m)\in\mathbb{C}^{I_m\times 1}$ denote the designed detector for user $i\in\mathcal{U}_m$ under the signal model specified in (16). Note that $\mathbf{H}_{\mathcal{I}_m}$ and $\bar{\mathbf{H}}_{\mathcal{I}_m}$ have been defined in Table II. The MSE of user i with $\bar{\mathbf{v}}_i(\mathcal{I}_m)$ is denoted by

$$\begin{aligned}\mathrm{MSE}\left(\bar{\mathbf{v}}_i(\mathcal{I}_m)\right)=&1+\left(\bar{\mathbf{v}}_i(\mathcal{I}_m)\right)^H\bar{\mathbf{R}}_{\mathcal{I}_m}\bar{\mathbf{v}}_i(\mathcal{I}_m)\\&-2\Re\left\{\sqrt{p_i}\left(\bar{\mathbf{v}}_i(\mathcal{I}_m)\right)^H\mathbf{H}_{\mathcal{I}_m}^H\mathbf{h}_i\right\},\end{aligned} \tag{17}$$

where $\bar{\mathbf{R}}_{\mathcal{I}_m}=\bar{\mathbf{H}}_{\mathcal{I}_m}^H\mathbf{\Lambda}\bar{\mathbf{H}}_{\mathcal{I}_m}+\sigma_z^2\mathbf{H}_{\mathcal{I}_m}^H\mathbf{H}_{\mathcal{I}_m}$. In particular, let $\bar{\mathbf{v}}_i^\star(\mathcal{I}_m)\in\mathbb{C}^{I_m\times 1}$ denote the MMSE detector for user i with (16). Then, we have the following proposition.

Proposition 2: For a given user set $\mathcal{U}_{m_1}\subseteq\mathcal{U}$ with $i\in\mathcal{U}_{m_1}$, let $\mathcal{U}_{m_2}=\mathcal{U}_{m_1}\cup\{\ell\}$, where $\ell\notin\mathcal{U}_{m_1}$, we always have

$$\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_2})\right)\le\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_1})\right), \tag{18}$$

where the equality holds if and only if

$$\mathbf{h}_\ell^H\mathbf{h}_j=0,\forall j\in\mathcal{U}. \tag{19}$$

Proof: The proof is available in Appendix B. □

From Proposition 2, the MSE performance of each UG can be improved by using the MF outputs of additional interfering users. Inspired by this observation, we propose to consider the users causing strong interference to $\mathcal{U}_m$ in MF process to improve the detection performance of $\mathcal{U}_m$. Therefore, we refer to $\mathcal{I}_m=\{i_1,i_2,\cdots,i_{I_m}\}$ as the interference group (IG) of UG $\mathcal{U}_m$. With the interference sparsity, as the users not in $\mathcal{I}_m$ cause little interference to $\mathcal{U}_m$, i.e., $\mathbf{H}_{\mathcal{I}_m}^H\mathbf{h}_j\approx 0, j\notin\mathcal{I}_m$, the matched filtered signal (16) for detecting $\mathcal{U}_m$ can be approximated by

$$\begin{aligned}\bar{\mathbf{y}}_{\mathcal{I}_m}=&\mathbf{H}_{\mathcal{I}_m}^H(\mathbf{H}_{\mathcal{I}_m}\mathbf{s}_{\mathcal{I}_m}+\mathbf{z})+\mathbf{H}_{\mathcal{I}_m}^H\sum_{j\notin\mathcal{I}_m}\mathbf{h}_j s_j\\\approx&\mathbf{H}_{\mathcal{I}_m}^H\tilde{\mathbf{y}}_{\mathcal{I}_m},\end{aligned} \tag{20}$$

where

$$\tilde{\mathbf{y}}_{\mathcal{I}_m}=\mathbf{H}_{\mathcal{I}_m}\mathbf{s}_{\mathcal{I}_m}+\mathbf{z}=\sum_{j\in\mathcal{U}_m}\mathbf{h}_j s_j+\sum_{j\in\mathcal{U}_m^{\mathcal{I}}}\mathbf{h}_j s_j+\mathbf{z}. \tag{21}$$

(20) is the reduced dimensional observation vector after MF, and (21) can be seen as the corresponding signal model before MF, where the interference outside $\mathcal{I}_m$ is omitted. The MMSE detector for detecting signals of $\mathcal{U}_m$ with (21) is given by

$$\begin{aligned}\mathbf{V}_{\mathcal{U}_m}^H=&\mathbf{\Lambda}_{\mathcal{U}_m}^{\frac{1}{2}}\mathbf{H}_{\mathcal{U}_m}^H\left(\mathbf{H}_{\mathcal{I}_m}\mathbf{\Lambda}_{\mathcal{I}_m}\mathbf{H}^H+\sigma_z^2\mathbf{I}_{BN}\right)^{-1}\\=&\mathbf{\Gamma}_m\left(\mathbf{\Lambda}_{\mathcal{I}_m}^{\frac{1}{2}}\mathbf{H}_{\mathcal{I}_m}^H\mathbf{H}_{\mathcal{I}_m}\mathbf{\Lambda}_{\mathcal{I}_m}^{\frac{1}{2}}+\sigma_z^2\mathbf{I}_{I_m}\right)^{-1}\mathbf{\Lambda}_{\mathcal{I}_m}^{\frac{1}{2}}\mathbf{H}_{\mathcal{I}_m}^H,\end{aligned} \tag{22}$$

where $\mathbf{\Gamma}_m=[\mathbf{I}_{\mathcal{U}_m},\mathbf{0}_{U_m\times(I_m-U_m)}]$ is a selection matrix. Obviously, $\mathcal{I}_m=U$ leads to the optimal detector (7). However in practice, incorporating all interfering users into each IG can present a considerable complexity challenge. Therefore, it is beneficial to exclude the users with little contribution to the MSE performance from the IG. For clarity, we let $\mathbf{v}_i^\star(\mathcal{I}_m)$ denote the MMSE detector designed for user $i\in\mathcal{U}_m$ with (22), and let $\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{I}_m)\right)$ denote the MSE of user i with $\mathbf{v}_i^\star(\mathcal{I}_m)$, which is given by

$$\mathrm{MSE}(\mathbf{v}_i^\star(\mathcal{I}_m))=1+(\mathbf{v}_i^\star(\mathcal{I}_m))^H\mathbf{R}\mathbf{v}_i^\star(\mathcal{I}_m)-2\Re\left\{\sqrt{p_i}\mathbf{h}_i^H\mathbf{v}_i^\star(\mathcal{I}_m)\right\} \tag{23}$$

with $\mathbf{R}=\mathbf{H}\mathbf{\Lambda}\mathbf{H}^H+\sigma_z^2\mathbf{I}$. Then, we have the following proposition.

Proposition 3: Let $\mathcal{U}_{m_2}=\mathcal{U}_{m_1}\cup\{\ell\}$ with $i\in\mathcal{U}_{m_1},\ell\notin\mathcal{U}_{m_1}$. If

$$\|\mathbf{h}_j^H\mathbf{h}_\ell\|\le\delta,\forall j\in\mathcal{U}_{m_1}, \tag{24}$$

there exists a constant $c>0$ such that the following holds

$$\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{U}_{m_1})\right)-\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{U}_{m_2})\right)\le c\delta. \tag{25}$$

Further, we have

$$\lim_{\delta\to 0}\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{U}_{m_2})\right)=\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{U}_{m_1})\right). \tag{26}$$

Proof: The proof is available in Appendix C. □

Proposition 3 presents an upper bound of the MSE performance loss of user i when the interference from a particular user is disregarded. According to Proposition 3, omitting the users with small interference with all the users in the UG results in negligible performance loss. As $\bar{\mathbf{H}}$ is inherently a numerically sparse matrix. Leveraging the interference sparsity, Proposition 3 implies that most users in the system just cause minimal interference for each UG, and the detection performance of the MMSE detector (22) can be guaranteed by only using the MF outputs of the intra-group users and a limited number of interfering users.

C. MMSE Detection With Both Interference Sparsity and BS Connection Sparsity

The signal model (21) for detecting signals of $\mathcal{U}_m$ can be written in a more detailed form as

$$\tilde{\mathbf{y}}_{\mathcal{I}_m}=\begin{bmatrix}\mathbf{h}_{i_1,1}&\cdots&\mathbf{h}_{i_{I_m},1}\\\mathbf{h}_{i_1,2}&\cdots&\mathbf{h}_{i_{I_m},2}\\\vdots&\ddots&\vdots\\\mathbf{h}_{i_1,B}&\cdots&\mathbf{h}_{i_{I_m},B}\end{bmatrix}\begin{bmatrix}s_{i_1}\\s_{i_2}\\\vdots\\s_{i_{I_m}}\end{bmatrix}+\begin{bmatrix}\mathbf{z}_1\\\mathbf{z}_2\\\vdots\\\mathbf{z}_B\end{bmatrix}. \tag{27}$$

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Note that the vectors $\mathbf{h}_{j,k}$ in $\mathbf{H}_{\mathcal{I}_m}$ can reflect the channel strength between users in $\mathcal{I}_m$ and all BSs. To be specific, when $\|\mathbf{h}_{j,k}\|$ is small, it is expected that BS k has weak channel connectivity with user j, and the received signal of BS k bears little information of s_j. In network mMIMO systems, the magnitude of each element in $\mathbf{h}_{j,k}$ tends to be small for BSs located distant form user j. This suggests that $\mathbf{h}_j$ is inherently a numerically block sparse vector, and the number of BSs with strong channel connectivity with each user is limited. Therefore, we refer to this property as the *BS connection sparsity*. With this sparsity, the BSs that have strong channel gains with user i constitute the associated BSs of user i denoted by $\bar{\mathcal{B}}_i$. On this basis, the channel $\mathbf{H}_{\mathcal{I}_m}$ is also expected to exhibit the block sparsity, indicating that the number of BSs that have strong channel connectivity with the users in $\mathcal{I}_m$ is limited. This suggests the possibility of ignoring certain BSs' received signals during detection for each UG without significant performance degradation.

Let $\mathcal{B}_m$ denote the set of BSs whose received signals are involved in detecting the signal of the users in $\mathcal{U}_m$ with $\mathrm{card}\,(\mathcal{B}_m) = B_m$. Specifically, $\mathcal{B}_m$ is termed as the associated BS cluster (BC) of the m-th UG. With the interference and BS connection sparsities, the observation vector of $\mathcal{B}_m$ after MF for detecting $\mathcal{U}_m$ is given by

$$\begin{aligned}\bar{\mathbf{y}}_{\mathcal{I}_m,\mathcal{B}_m}&=\mathbf{H}^H_{\mathcal{I}_m,\mathcal{B}_m}\mathbf{y}_{\mathcal{B}_m}\\&=\mathbf{H}^H_{\mathcal{I}_m,\mathcal{B}_m}(\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m}\mathbf{s}_{\mathcal{I}_m}+\mathbf{z}_{\mathcal{B}_m})+\mathbf{H}^H_{\mathcal{I}_m,\mathcal{B}_m}\sum_{j\notin\mathcal{I}_m}\mathbf{h}_{j,\mathcal{B}_m}s_j\\&\approx\mathbf{H}^H_{\mathcal{I}_m,\mathcal{B}_m}(\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m}\mathbf{s}_{\mathcal{I}_m}+\mathbf{z}_{\mathcal{B}_m})=\mathbf{H}^H_{\mathcal{I}_m,\mathcal{B}_m}\tilde{\mathbf{y}}_{\mathcal{I}_m,\mathcal{B}_m}\end{aligned}\tag{28}$$

with

$$\tilde{\mathbf{y}}_{\mathcal{I}_m,\mathcal{B}_m}=\mathbf{H}_{\mathcal{U}_m,\mathcal{B}_m}\mathbf{s}_{\mathcal{U}_m}+\mathbf{H}_{\mathcal{U}^{\mathcal{I}}_m,\mathcal{B}_m}\mathbf{s}_{\mathcal{U}^{\mathcal{I}}_m}+\mathbf{z}_{\mathcal{B}_m},\tag{29}$$

where $\bar{\mathbf{y}}_{\mathcal{I}_m,\mathcal{B}_m}\in\mathbb{C}^{I_m\times 1}$ represents the reduced dimensional vector after MF with interference and BS connection sparsities, and (29) can be seen as the corresponding signal model for detecting signals of $\mathcal{U}_m$. $\mathbf{z}_{\mathcal{B}_m}$ is the stacked noise. Unlike the signal model in (3), the dimension of the observation vector and the number of interfering users are all reduced in (29). Then, the MMSE detector for detecting signals of $\mathcal{U}_m$ with (29) takes the form of

$$\begin{aligned}\mathbf{V}^H_{\mathcal{U}_m,\mathcal{B}_m}&=\mathbf{\Lambda}^{\frac{1}{2}}_{\mathcal{U}_m}\mathbf{H}^H_{\mathcal{U}_m,\mathcal{B}_m}\left(\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m}\mathbf{\Lambda}_{\mathcal{I}_m}\mathbf{H}^H_{\mathcal{I}_m,\mathcal{B}_m}+\sigma_z^2\mathbf{I}_{B_mN}\right)^{-1}\\&=\mathbf{\Gamma}_m\left(\mathbf{\Lambda}^{\frac{1}{2}}_{\mathcal{I}_m}\mathbf{H}^H_{\mathcal{I}_m,\mathcal{B}_m}\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m}\mathbf{\Lambda}^{\frac{1}{2}}_{\mathcal{I}_m}+\sigma_z^2\mathbf{I}\right)^{-1}\mathbf{\Lambda}^{\frac{1}{2}}_{\mathcal{I}_m}\mathbf{H}^H_{\mathcal{I}_m,\mathcal{B}_m},\end{aligned}\tag{30}$$

where $\mathbf{\Gamma}_m$ has been defined in (22). For brevity and clarity, let $\mathbf{v}_i^\star(\mathcal{I}_m,\mathcal{B}_m)$ denote the MMSE detector for user $i\in\mathcal{U}_m$ with (30) that only considers $\mathcal{I}_m$ and $\mathcal{B}_m$ as the IG and BC, respectively. The MSE performance of user i with $\mathbf{v}_i^\star(\mathcal{I}_m,\mathcal{B}_m)$ is given by

$$\begin{aligned}&\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{I}_m,\mathcal{B}_m)\right)\\&=1-p_i\mathbf{h}^H_{i,\mathcal{B}_m}\left(\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m}\mathbf{\Lambda}_{\mathcal{I}_m}\mathbf{H}^H_{\mathcal{I}_m,\mathcal{B}_m}+\sigma_z^2\mathbf{I}_{B_mN}\right)^{-1}\mathbf{h}_{i,\mathcal{B}_m}.\end{aligned}\tag{31}$$

Then, we have the following proposition.

Proposition 4: Let $\mathcal{B}_{m_1}\subseteq\mathcal{B}$ and $\mathcal{B}_{m_2}=\mathcal{B}_{m_1}\cup\{\ell\}$ with $\ell\notin\mathcal{B}_{m_1}$, we always have

$$\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{I}_m,\mathcal{B}_{m_2})\right)\le\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{I}_m,\mathcal{B}_{m_1})\right),\tag{32}$$

where the equation holds if and only if $\mathbf{H}_{\mathcal{I}_m,\ell}=\mathbf{0}$. The performance gap can be further quantified by

$$\mathrm{MSE}(\mathbf{v}_i^\star(\mathcal{I}_m,\mathcal{B}_{m_2}))=\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{I}_m,\mathcal{B}_{m_1})\right)-\zeta_{m,\ell},\tag{33}$$

where $\zeta_{m,\ell}=p_i\mathbf{h}^H_{i,\mathcal{B}_{m_2}}\mathbf{B}\mathbf{h}_{i,\mathcal{B}_{m_2}}$ with $\mathbf{B}$ being a semi-positive matrix.

Proof: The proof and detailed $\mathbf{B}$ are available in Appendix D. □

From (32) in Proposition 4, the MSE performance of each UG will never decrease when additional BSs are included in the detection process, and the optimal performance will be achieved when all BSs in the system are considered. However, (32) also indicates that incorporating the received signal of the ℓ-th BS into detection will not improve the MSE performance of $\mathcal{U}_m$ if $\mathbf{H}_{\mathcal{I}_m,\ell}=\mathbf{0}$. In this case, the ℓ-th BS can be omitted during detection without loss of optimality. This result reveals that only the BS that have no association with all users in $\mathcal{I}_m$ can be omitted without performance loss. In other words, the performance-preserving BC of $\mathcal{U}_m$ corresponds to the union of the associated BSs of all users in $\mathcal{I}_m$, i.e., $\mathcal{B}_m=\bigcup_{i\in\mathcal{I}_m}\bar{\mathcal{B}}_i$.

In practice, achieving $\mathbf{H}_{\mathcal{I}_m,\ell}=\mathbf{0}$ is unattainable. To reduce the complexity and backhaul burden, it is beneficial to exclude BSs that make little contribution to the MSE performance of each UG from its BC. In this regard, (33) is provided to quantify the accurate MSE gap resulting from the exclusion of a given BS from the BC. However, the computation of $\zeta_{m,\ell}$ is complicated, making practical application difficult. In order to establish a more feasible criterion for constructing the BC $\mathcal{B}_m$, the following proposition is presented.

Proposition 5: Let $\mathcal{B}_{m_2}=\mathcal{B}_{m_1}\cup\{\ell\}$ with $\ell\notin\mathcal{B}_{m_1}$. If $\|\mathbf{H}_{\mathcal{I}_m,\ell}\|_\infty\le\phi$, there exists a constant $p>0$ such that the following holds

$$\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{I}_m,\mathcal{B}_{m_1})\right)-\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{I}_m,\mathcal{B}_{m_2})\right)\le p\phi.\tag{34}$$

Further, we have

$$\lim_{\phi\to 0}\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{I}_m,\mathcal{B}_{m_2})\right)=\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{I}_m,\mathcal{B}_{m_1})\right).\tag{35}$$

Proof: The proof is available in Appendix E. □

Proposition 5 presents an upper bound for the MSE performance loss when the received signal from a certain BS is omitted during the detection process for $\mathcal{U}_m$. From Proposition 5, if the norm of each element in $\mathbf{H}_{\mathcal{I}_m,\ell}$is negligible, BS ℓ can be excluded from the BC $\mathcal{B}_m$ during the signal detection for $\mathcal{U}_m$ without much performance loss. In contrast to the accurate performance gap $\zeta_{m,\ell}$ derived in Proposition 4, $\|\mathbf{H}_{\mathcal{I}_m,\ell}\|_\infty$ is more accessible and computationally efficient, making it more applicable in practice. As discussed previously, $\mathbf{H}_{\mathcal{I}_m}$ is inherently a block sparse matrix, and the condition for (35) is inherently satisfied for the BSs located distant from the IG $\mathcal{I}_m$. With the BS connection sparsity, we reveal that the detection performance of each UG is determined by a small

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set of BSs in the system, allowing for a reduction in received signal dimensions without much loss of performance.

With Proposition 3 and Proposition 5, we show that the MMSE detector (30) can approach the optimal detection performance as (7) with the interference and BS connection sparsities. As the number of interfering users for $\mathcal{U}_m$ is limited, it is typically expected that $U_m \leq I_m < B_m N$. In this way, the design and implementation complexities can be reduced to $O\left(\sum_{m=1}^{M}\left(B_m N I_m^2 + B_m N I_m U_m + I_m^3\right)\right)$ and $\sum_{m=1}^{M} B_m N U_m$, respectively.

III. Signal Detection for UCN mMIMO System

With the inherent interference sparsity and BS connection sparsity, we have demonstrated that the performance of the MMSE detection can be guaranteed by only considering a small number of interfering users and a small subset of BSs per UG. However, the detection still involves the cooperation of the BSs in each BC. Generally, a central processing unit (CPU) is required for centralized processing, and all BSs need to exchange information with the CPU, making the system unscalable. Considering that it is sufficient to exchange information among the BSs in each BC during detection for each UG, each UG can be assigned a CPU for detector design to avoid global information exchange. Nevertheless, this indicates that the number and placement of CPUs are determined by UGs, which change dynamically over time, presenting challenges for its practical realization. To address these issues, we present a user-centric detection approach for UCN mMIMO systems in this section, where the primary served users (PSUs) of each BS constitute a UG and the signal detection of each UG is performed at its PSBS, with necessary information exchanged with the ASBSs.

A. User Grouping and BC/IG Construction for UCN mMIMO System

Before introducing the user-centric MMSE detection, we first clarify how the UCN mMIMO system works. Specifically, we present an efficient approach for user grouping, and provide the construction criteria of the BC and IG for each UG.

In the proposed UCN mMIMO system, each UG experiences interference from a limited number of interfering users and served by a subset of BSs, with one serving as the PSBS and others acting as the ASBSs. Each PSBS is responsible for serving a group of users by locally exchanging the required information with the corresponding ASBSs.

First of all, we present a method for determining the PSBS and associated BSs of each user. In practice, the associated BSs of each user can be identified according to the large-scale fading or the reference signal received power (RSRP). Let

$$\phi_{i,k} = \mathbb{E}\left\{\mathbf{h}_{i,k}^H \mathbf{h}_{i,k}\right\}, \tag{36}$$

which quantifies the mean channel power between user i and BS k. Then, the BS with the best channel conditions for user i can be defined as

$$k_i^\star = \arg\max_{k\in\mathcal{B}} \phi_{i,k}, \tag{37}$$

and is referred to as the PSBS of user i. With Proposition 5, we can define

$$\kappa_{i,\ell} = \phi_{i,k_i^\star} - \phi_{i,\ell} \tag{38}$$

to measure the difference in channel gain experienced by user i from its PSBS relative to BS ℓ. Then, the associated BS set $\bar{\mathcal{B}}_i$ of user i can be constructed by

$$\bar{\mathcal{B}}_i = \left\{\ell \in \mathcal{B} | \kappa_{i,\ell} < \delta_{cbs}^i\right\}, \tag{39}$$

where δ_{cbs}^i is the threshold. In this way, the BS with strong channel connectivity with each user can be included in the associated BS set, whose cardinality can be adjusted by varying the threshold δ_{cbs}^i. Note that (36) varies relatively slow and the results can be reused for a period of time [46], [47]. Therefore, this process incurs negligible costs. Once the associated BSs of each user is determined, the served users of the k-th BS are known and denoted by $\bar{\mathcal{U}}_k$. In practice, the users can scan the synchronization signal blocks (SSB) of nearby BSs, and measure the signal strength from each of them, thereby determining the PSBS and associated BSs [48]. For clarity, the users whose PSBS is k are termed as the PSUs of BS k and the set is denoted by $\mathcal{U}_k^{\mathrm{P}}$, and the rest users served by BS k are termed as the secondary served users (SSUs), and the user set is denoted by $\mathcal{U}_k^{\mathrm{S}} = \bar{\mathcal{U}}_k \backslash \mathcal{U}_k^{\mathrm{P}}$.

After each user has identified the PSBS, a straightforward and efficient method for user grouping is to let the PSUs of BS k be the k-th UG, i.e., let $\mathcal{U}_k = \mathcal{U}_k^{\mathrm{P}}$. Note that $\mathcal{U}_k$ could be an empty set. As discussed in Section II-B, the users with the same PSBS tend to be geographically closer, and are expected to have relatively strong interference with each other. In this way, the signals for each UG can naturally be processed at their respective PSBSs with necessary information exchanged from ASBSs, which offers improved scalability.

Then, we clarify how to determine the IG for each UG. Let $\tilde{\mathcal{B}}_k = \cup_{i\in\mathcal{U}_k} \bar{\mathcal{B}}_i$ denote the union set of associated BSs associated with the PSUs of BS k, recalling that $\mathcal{U}_k = \bar{\mathcal{U}}_k^{\mathrm{P}}$ is the k-th UG and $\bar{\mathcal{B}}_i$ is the associated BS set of user i. As the BSs in $\tilde{\mathcal{B}}_k$ locate near the users in $\mathcal{U}_k$, it is expected that their served users are more likely to cause relatively strong interference to the users in $\mathcal{U}_k$. Therefore, we propose to select the interfering users from $\cup_{\ell\in\tilde{\mathcal{B}}_k} \mathcal{U}_\ell$ for $\mathcal{U}_k$ rather than from all users. Note that the IG $\mathcal{I}_k$ can be initialized by $\mathcal{U}_k$. Let

$$\psi_{k,j,\ell} = \frac{1}{U_k} \sum_{i\in\mathcal{U}_k} \frac{\|\mathbf{h}_{j,\ell}^H \mathbf{h}_{i,\ell}\|}{\|\mathbf{h}_{i,\ell}^H \mathbf{h}_{i,\ell}\|}, \ell \in \tilde{\mathcal{B}}_k, j \in \mathcal{U}_\ell. \tag{40}$$

If

$$\psi_{k,j,\ell} > \delta_{ig}^k, \tag{41}$$

where $\delta_{ig}^k > 0$, then, j can be included in $\mathcal{I}_k$, otherwise, j is excluded from $\mathcal{I}_k$. The users with strong interference will be included in the IG of each UG, and the size of IG can be adjusted with different δ_{ig}^k. As user j in (40) belongs to the served users $\mathcal{U}_\ell$ of BS ℓ, (40) can be obtained with the instantaneous CSI estimated at the pilot segments, and the resulting IG can be reused over a period time when the mobility is relatively low.

Once the associated BSs of each user and the IG of each UG are determined, the BC of $\mathcal{U}_k$ can be obtained by

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$\mathcal{B}_k = \bigcup_{i\in\mathcal{I}_k} \bar{\mathcal{B}}_i$ with Proposition 4. In particular, the BSs in $\mathcal{B}_k \backslash k$ are referred to as the ASBSs of the k-th UG, and the BS set is denoted by $\mathcal{B}_k^{\mathrm{A}}$ with $\mathrm{card}\left(\mathcal{B}_k^{\mathrm{A}}\right) = B_k^{\mathrm{A}}$. In this way, each UG can be served by its PSBS with necessary information exchanged with BSs in $\mathcal{B}_k^{\mathrm{A}}$ in our proposed UCN mMIMO system. The computational complexity of our proposed BC construction criterion is $\mathrm{O}(NBU)$, which is comparable with the largest effective channel gain (LECG) criterion proposed in [46]. Additionally, the computational complexities of large-scale fading decoding (LSFD) based criterion proposed in [49], and the Log-Likelihood Mean (LLR-M) criterion proposed in [47], are $O\left(\sum_{i\in\mathcal{U}} U^4 B_i^2\right)$ and $\mathrm{O}\left(\sum_{i\in\mathcal{U}}(U^4B_i^2 + B_i^3) + BN^3\right)$, respectively, which are all much higher than that of our proposed BC construction criterion.

Compared to the conventional network mMIMO system, UCN mMIMO system inherently works in a more distributed manner. Specifically, the signal detection for each UG in the UCN mMIMO system can be performed separately at respective PSBS, with only a small subset of users considered as the interfering users and a limited number of BSs as the associated BSs. The information only needs to be locally exchanged between the PSBS and ASBSs. Note that the UCN system is reduced to the traditional cellular mMIMO system if there is only one BS in each BC, and the system becomes the conventional network mMIMO system when $\mathcal{B}_k = \mathcal{B}$ and $\mathcal{I}_k = U$. More importantly, Proposition 2 and Proposition 4 indicate that the MMSE detection in the proposed UCN mMIMO system can achieve the optimal performance under certain conditions, demonstrating its effectiveness and optimality. Further, Proposition 3 and Proposition 5 ensure that the detection performance of each UG in UCN mMIMO systems can still approach the optimal performance by discarding some users and BSs with little contribution, thereby reducing the computational complexity and facilitating practical application.

B. User-Centric Signal Detection

In this subsection, we introduce the user-centric signal detection method for UCN mMIMO system. For each UG, the proposed method decouples the detection process into two stages: the first stage is local MF at the baseband unit (BBU) of each associated BS, and the second stage is group-wise IC at the BBU of PSBS with necessary information exchanged from the ASBSs.

We focus on the signal detection for the k-th UG. With the UG, BC and IG determined in Section III-A, the observation vector of $\mathcal{B}_k$ after MF for detecting $\mathcal{U}_k$ is given by

$$\begin{aligned}\bar{\mathbf{y}}_{\mathcal{I}_k,\mathcal{B}_k} &= \mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H \mathbf{y}_{\mathcal{B}_k} \\ &= \mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H(\mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}\mathbf{s}_{\mathcal{I}_k} + \mathbf{z}_{\mathcal{B}_k}) + \mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H \sum_{j\notin\mathcal{I}_k} \mathbf{h}_{j,\mathcal{B}_k} s_j \\ &\approx \mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H(\mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}\mathbf{s}_{\mathcal{I}_k} + \mathbf{z}_{\mathcal{B}_k}) = \mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H \tilde{\mathbf{y}}_{\mathcal{I}_k,\mathcal{B}_k} \end{aligned} \tag{42}$$

with

$$\tilde{\mathbf{y}}_{\mathcal{I}_k,\mathcal{B}_k} = \mathbf{H}_{\mathcal{U}_k,\mathcal{B}_k}\mathbf{s}_{\mathcal{U}_k} + \mathbf{H}_{\mathcal{U}_k^{\mathcal{I}},\mathcal{B}_k}\mathbf{s}_{\mathcal{U}_k^{\mathcal{I}}} + \mathbf{z}_{\mathcal{B}_k}. \tag{43}$$

$\tilde{\mathbf{y}}_{\mathcal{I}_k,\mathcal{B}_k} \in \mathbb{C}^{B_kN\times 1}$ can be seen as the signal model with interference and BS connection sparsities for detecting signals of $\mathcal{U}_k$. Let $\mathbf{V}_{\mathcal{U}_k,\mathcal{B}_k}^H$ denote the detector for detecting signals of $\mathcal{U}_k$ with $\tilde{\mathbf{s}}_{\mathcal{U}_k} = \mathbf{V}_{\mathcal{U}_k,\mathcal{B}_k}^H \tilde{\mathbf{y}}_{\mathcal{I}_k,\mathcal{B}_k}$. Then, the MSE performance of the k-th UG is given by

$$\begin{aligned}&\mathrm{MSE}_k^{\mathrm{UCN}} \\ &= \mathbb{E}\left\{\left(\tilde{\mathbf{s}}_{\mathcal{U}_k} - \mathbf{\Lambda}_{\mathcal{U}_k}^{-\frac{1}{2}}\mathbf{s}_{\mathcal{U}_k}\right)^H \left(\tilde{\mathbf{s}}_{\mathcal{U}_k} - \mathbf{\Lambda}_{\mathcal{U}_k}^{-\frac{1}{2}}\mathbf{s}_{\mathcal{U}_k}\right) \mid \mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}\right\} \\ &= \mathrm{tr}\left(\mathbf{V}_{\mathcal{U}_k,\mathcal{B}_k}^H\left(\mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}\mathbf{\Lambda}_{\mathcal{I}_k}\mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H + \sigma_z^2\mathbf{I}_{B_kN}\right)\mathbf{V}_{\mathcal{U}_k,\mathcal{B}_k}\right) \\ &\quad - \mathrm{tr}\left(\mathbf{V}_{\mathcal{U}_k,\mathcal{B}_k}^H\mathbf{H}_{\mathcal{U}_k,\mathcal{B}_k}\mathbf{\Lambda}_{\mathcal{U}_k}^{\frac{1}{2}} + \mathbf{\Lambda}_{\mathcal{U}_k}^{\frac{1}{2}}\mathbf{H}_{\mathcal{U}_k,\mathcal{B}_k}^H\mathbf{V}_{\mathcal{U}_k,\mathcal{B}_k} - \mathbf{I}\right). \end{aligned} \tag{44}$$

Further, the user-centric detector that minimizes (44) takes the form of

$$\begin{aligned}\mathbf{V}_{\mathcal{U}_k,\mathcal{B}_k}^H &= \mathbf{\Gamma}_{\mathcal{U}_k}\left(\mathbf{\Lambda}_{\mathcal{I}_k}^{\frac{1}{2}}\mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H\mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}\mathbf{\Lambda}_{\mathcal{I}_k}^{\frac{1}{2}} + \sigma_z^2\mathbf{I}\right)^{-1}\mathbf{\Lambda}_{\mathcal{I}_k}^{\frac{1}{2}}\mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H \\ &= \mathbf{\Gamma}_{\mathcal{U}_k}\left(\mathbf{G}_{\mathcal{I}_k,k} + \sum_{j\in\mathcal{B}_k^{\mathrm{A}}}\mathbf{G}_{\mathcal{I}_k,j} + \sigma_z^2\mathbf{I}\right)^{-1}\mathbf{\Lambda}_{\mathcal{I}_k}^{\frac{1}{2}}\mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H \\ &= \underbrace{\bar{\mathbf{V}}_{\mathcal{U}_k,\mathcal{B}_k}^H}_{\mathrm{MMSE-IC}}\underbrace{\mathbf{\Lambda}_{\mathcal{I}_k}^{\frac{1}{2}}\left[\mathbf{H}_{\mathcal{I}_k,k_1}^H,\cdots,\mathbf{H}_{\mathcal{I}_k,k_{B_k}}^H\right]}_{\mathrm{Matched\ Filter}}, \end{aligned} \tag{45}$$

where $\mathbf{\Gamma}_{\mathcal{U}_k} \in \mathbb{C}^{U_k\times I_k}$ is a selection matrix with (i,x_i)-th element being 1 and the other elements being 0, where x_i is the index of user i in $\mathcal{I}_k$. $\mathbf{G}_{\mathcal{I}_k,j} \triangleq \mathbf{\Lambda}_{\mathcal{I}_k}^{1/2}\mathbf{H}_{\mathcal{I}_k,j}^H\mathbf{H}_{\mathcal{I}_k,j}\mathbf{\Lambda}_{\mathcal{I}_k}^{1/2} \in \mathbb{C}^{I_k\times I_k}$ is the channel Gram matrix with transmit power included. Recall that $\bar{\mathbf{V}}_{\mathcal{U}_k,\mathcal{B}_k}^H$ is the MMSE detector for the k-th UG using the MF outputs for IC. Therefore, we refer to $\bar{\mathbf{V}}_{\mathcal{U}_k,\mathcal{B}_k}^H$ as the group-wise MMSE-IC matrix. It can be seen from (45) that the detector $\mathbf{V}_{\mathcal{U}_k,\mathcal{B}_k}^H$ is composed of the group-wise matched filter and the MMSE-IC matrix. With (45), the signal detection for $\mathcal{U}_k$ can be completed by

$$\begin{aligned}\hat{\mathbf{s}}_{\mathcal{U}_k} &= \mathbf{V}_{\mathcal{U}_k,\mathcal{B}_k}^H\mathbf{y}_{\mathcal{B}_k} \\ &= \underbrace{\bar{\mathbf{V}}_{\mathcal{U}_k,\mathcal{B}_k}^H}_{\mathrm{IC}}\sum_{j\in\mathcal{B}_k}\underbrace{\mathbf{\Lambda}_{\mathcal{I}_k}^{\frac{1}{2}}\mathbf{H}_{\mathcal{I}_k,j}^H\mathbf{y}_j}_{\mathrm{Local\ MF}} = \bar{\mathbf{V}}_{\mathcal{U}_k,\mathcal{B}_k}^H\sum_{\ell\in\mathcal{B}_k}\bar{\mathbf{y}}_{\mathcal{I}_k,\ell}, \end{aligned} \tag{46}$$

where $\bar{\mathbf{y}}_{\mathcal{I}_k,\ell}$ represents the local MF outputs of BS ℓ for detecting $\mathcal{U}_k$. It can be seen that the signal detection for $\mathcal{U}_k$ is completed byM local MF followed by group-wise IC with MMSE-IC matrix. Note that the local MF outputs are available to each BS with local CSI and received signal as shown in (46). However, further IC process requires information exchange between BSs. To be specific, the group-wise MMSE-IC matrix computation and signal detection are performed at the PSBS for each UG with information exchanged from ASBSs. For the k-th UG, the group-wise MMSE-IC matrix $\bar{\mathbf{V}}_{\mathcal{U}_k,\mathcal{B}_k}^H$ computation and the signal detection requires $\mathbf{G}_{\mathcal{I}_k,j}$ and $\bar{\mathbf{y}}_{\mathcal{U}_k,j}, j \in \mathcal{B}_k^{\mathrm{A}}$, respectively, as shown in (45) and (46). In this way, the proposed user-centric detection for UCN mMIMO system decouples the detection process into two stages for each UG:

1) **Local MF:** Each associated BS performs MF and computes the channel Gram matrix locally using the channels of intra-group and interfering users.
2) **Group-wise IC:** PSBS computes the group-wise MMSE-IC matrix with the channel Gram matrices

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TABLE IV
COMPARISON OF COMPUTATIONAL COMPLEXITY AND BACKHAUL BURDEN FOR DIFFERENT DETECTORS

	Optimal Detector	User-centric Detector
Design Complexity	$2BNU^2+U^3$	$\sum_{k=1}^{B}\left(B_kNI_k^2+B_kNI_kU_k+I_k^3\right)$
Implementation Complexity	BNU	$\sum_{k=1}^{B}B_kNU_k$
Backhaul burden	BNU	$\sum_{k\in\mathcal{B}}\sum_{\ell\in\mathcal{B}_k^{\mathrm{P}}}\left(I_\ell^2+I_\ell\right)$

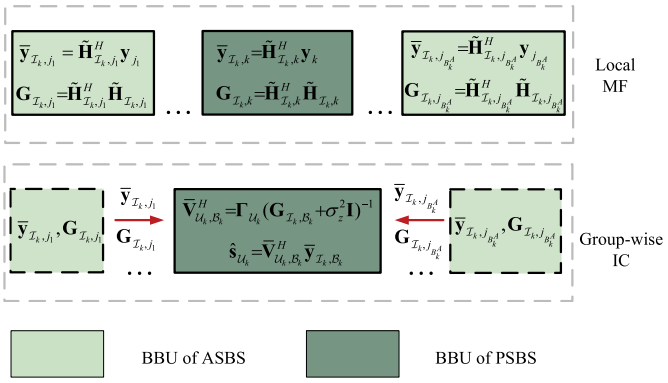


Fig. 1. The computations and information exchange during the user-centric detection for the k-th UG.

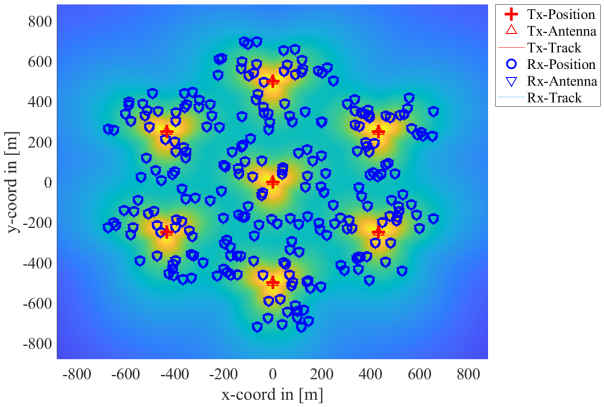


Fig. 2. An illustration of the system layout.

exchanged from ASBSs. Then, PSBS applies the MMSE-IC matrix to the sum of local MF outputs for final detection results.

An illustration of the user-centric detection for the k-th UG is shown in Fig. 1, where $\tilde{\mathbf{H}}_{\mathcal{I}_k,j}=\mathbf{\Lambda}_{\mathcal{I}_k}^{1/2}\mathbf{H}_{\mathcal{I}_k,j}^H$, $\mathbf{G}_{\mathcal{I}_k,\mathcal{B}_k}=\sum_{j\in\mathcal{B}_k}\mathbf{G}_{\mathcal{I}_k,j}$ and $\bar{\mathbf{y}}_{\mathcal{I}_k,\mathcal{B}_k}=\sum_{j\in\mathcal{B}_k}\bar{\mathbf{y}}_{\mathcal{I}_k,j}$. The user-centric nature of our proposed detection is reflected in the individual detection and localized information requirements of each UG. For each UG, the local MF and MMSE-IC matrix only require the channels from the intra-group and interfering users to the associated BSs. The user-centric detector presented in (45) can be seen as a generic detector with different configurations of IG and BC. Specifically, if $\mathcal{I}_k=\mathcal{U}$ and $\mathcal{B}_k=\mathcal{B}$, (45) becomes the optimal detector (7) for the network mMIMO system. By setting $\mathcal{I}_k=\mathcal{U}_k$ and $\mathcal{B}_k=\{k\}$, (45) is reduced to the MMSE detector for the conventional cellular mMIMO system.

The backhaul load of the k-th BS during the user-centric detection is $\sum_{\ell\in\mathcal{B}_k^{\mathrm{P}}}\left(I_\ell^2+I_\ell\right)$, where $\mathcal{B}_k^{\mathrm{P}}=\cup_{i\in\mathcal{U}_k^{\mathrm{S}}}k_i^\star$ denotes the set of the PSBSs associated with the SSUs of BS k. Note that the backhaul burden is independent of the number of antennas and the total number of BSs, and is only related to the number of interfering users of each UG. The design and implementation complexities of the user-centric detector (45) for all UGs are $O\left(\sum_{k=1}^{B}\left(B_kNI_k^2+B_kNI_kU_k+I_k^3\right)\right)$ and $\sum_{k=1}^{B}B_kNU_k$, respectively. For clarity, we provide a comprehensive comparison of the complexity and link load between the proposed user-centric detector and the optimal MMSE detector in Table IV.

IV. SIMULATION RESULTS

In this section, we provide simulation results to validate the superiority of our proposed user-centric detection for UCN mMIMO system.

TABLE V
DETAILED SIMULATION PARAMETERS

Center frequency	6.725 GHz	U	256
BS antenna type	16×8 UPA	σ_z^2	-112 dBW
Modulation Mode	QAM	Modulation Order	4, 16

We adopt the prevalent QuaDRiGa channel model [50] to generate a simulation scenario, where "3GPP 38.901 RMa NLOS" is considered. To ensure a better coverage, we consider the tri-sector configuration and seven gNodeBs (gNBs) are installed in the system [51]. Each gNB has three co-located BSs and each BS is responsible for a 120-degree coverage [52] as shown in Fig. 2. So there are totally $B=21$ BSs in the system. The distance between the adjacent gNBs is set to 500 m in our simulations [53]. In the network, the users are distributed in a circle with radius of 750 m. The UG, IG and BC are determined via the approaches introduced in Section III-A. The transmit power of each user is assumed the same and is denoted by p, which is set according to [54]. More detailed system parameters are summarized in Table V. For ease of comparison, we let $\bar{B}_{\mathrm{cbs}}$ denote the number of associated BSs of each user and let I_{ig} denote the number of interfering users of each UG. For brevity, the user-centric detector in (45) that considers I_{ig} interfering users and $\bar{B}_{\mathrm{cbs}}$ associated BSs is termed as UCD with I_{ig} and $\bar{B}_{\mathrm{cbs}}$. Recall that UCD with $I_{\mathrm{ig}}=U$ and $\bar{B}_{\mathrm{cbs}}=B$ leads to the optimal MMSE detector for the network mMIMO system in (7), and UCD with $I_{\mathrm{ig}}=U_k$ and $\bar{B}_{\mathrm{cbs}}=1$ becomes the MMSE detector for the traditional cellular mMIMO system. Therefore, the UCD with $I_{\mathrm{ig}}=U$ and $\bar{B}_{\mathrm{cbs}}=B$ is referred to as

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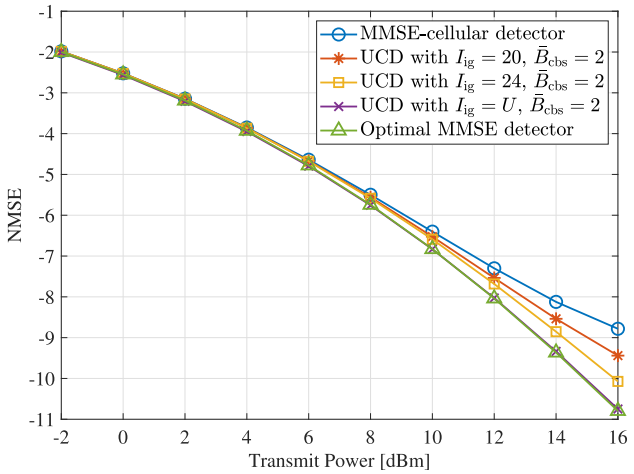


Fig. 3. NMSE performance of user-centric detector under 4-QAM.

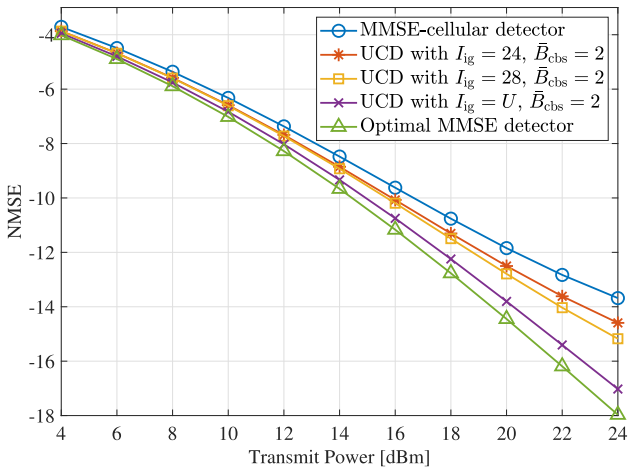


Fig. 4. NMSE performance of user-centric detector under 16-QAM.

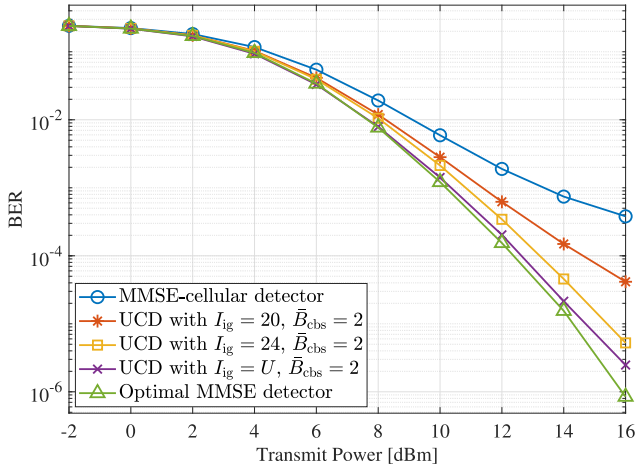


Fig. 5. BER performance of user-centric detector under 4-QAM.

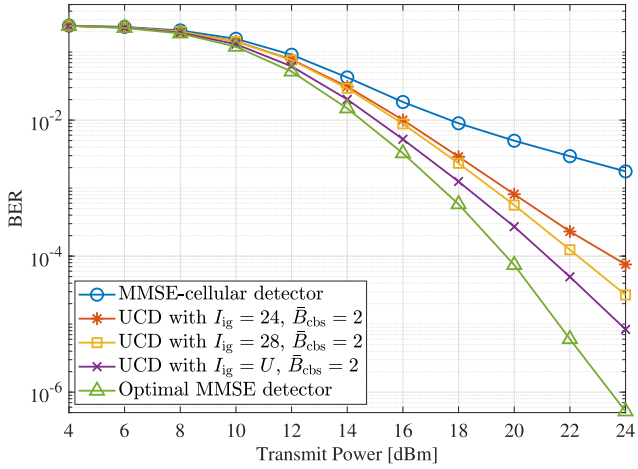


Fig. 6. BER performance of user-centric detector under 16-QAM.

the optimal MMSE detector, and UCD with $I_{\text{ig}} = U_k$ and $\bar{B}_{\text{cbs}} = 1$ is termed as the MMSE-cellular detector for brevity. In the following, we set $\bar{B}_{\text{cbs}} = 2$ for UCD, which can lead to satisfactory performance according to [40].

First of all, we provide simulation results to show the normalized mean-squared error (NMSE) performance of the proposed user-centric detection for UCN mMIMO system. We can see from Fig. 3 that the UCD with $I_{\text{ig}} = 24$ and $\bar{B}_{\text{cbs}} = 2$ has little NMSE performance loss compared with the UCD with $I_{\text{ig}} = U$ and $\bar{B}_{\text{cbs}} = 2$, which shares nearly the same performance as the optimal MMSE detector under 4-QAM modulation, demonstrating that the NMSE performance can be guaranteed by only considering a limited number of interfering users and associated BSs. For 16-QAM, it can be seen from Fig. 4 that the UCD with $I_{\text{ig}} = 24$ and $\bar{B}_{\text{cbs}} = 2$ has nearly the same NMSE performance as the UCD with $I_{\text{ig}} = 28$ and $\bar{B}_{\text{cbs}} = 2$, and has only about 3 dB performance loss compared with the optimal MMSE detector when $p = 24$ dBm. Additionally, for both modulation schemes, the UCD with $I_{\text{ig}} = 24$ and $\bar{B}_{\text{cbs}} = 2$ has a better NMSE performance compared with the MMSE-cellular detector with $I_{\text{ig}} = U_k$ and $\bar{B}_{\text{cbs}} = 1$.

Fig. 5 and Fig. 6 show the bit error rate (BER) performance of the user-centric detection for UCN mMIMO system. For 4-QAM modulation scheme, it can be seen from Fig. 5 that the UCD with $I_{\text{ig}} = 24$ and $\bar{B}_{\text{cbs}} = 2$ has demonstrated satisfactory BER performance. Specifically, it has nearly the same BER performance as the optimal MMSE detector for the network mMIMO system in the medium transmit power regime, and has only about 1 dB performance loss compared with the optimal MMSE detector when $p = 16$ dBm, demonstrating that the BER performance can be guaranteed by considering a limited number of interfering users and associated BSs. For 16-QAM, the UCD with $I_{\text{ig}} = 24$ and $\bar{B}_{\text{cbs}} = 2$ has nearly the same BER performance as the UCD $I_{\text{ig}} = 28$ and $\bar{B}_{\text{cbs}} = 2$. In addition, it has only about 2 dB performance loss compared with $I_{\text{ig}} = U$ and $\bar{B}_{\text{cbs}} = 2$ when $p = 22$ dBm. More importantly, the UCD with $I_{\text{ig}} = 24$ and $\bar{B}_{\text{cbs}} = 2$ has a much better BER performance compared with the MMSE-cellular detector.

Then, we provide a performance comparison across different BS selection methods, including the LECG criterion proposed in [46], the LSFD based criterion proposed in [49], and LLR-M criterion proposed in [47]. For fair of comparison, we employ the proposed user-centric detector while solely

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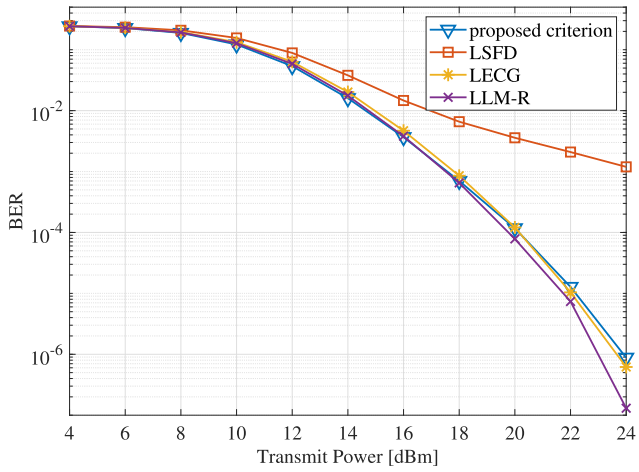


Fig. 7. BER performance under different BS selection criteria.

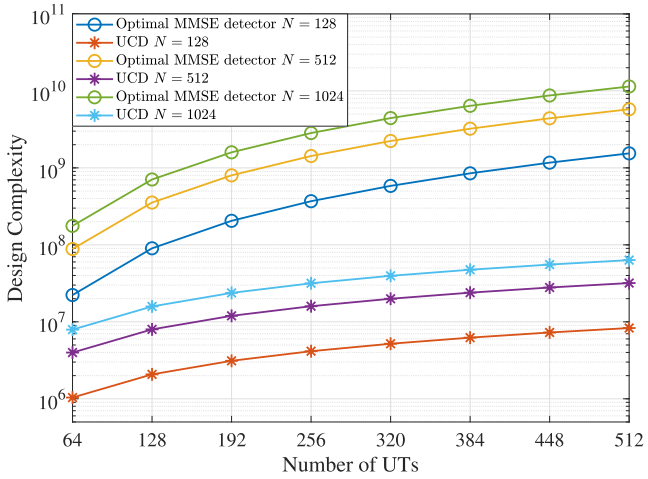


Fig. 8. Comparison of design complexities between the user-centric detector and the optimal MMSE detector.

varying the BS selection strategy. As shown in Fig. 7, we can see that our proposed criterion share similar performance as LECG, while outperforms LSFD. This is because that in our approach, the BC of each UG is the union of all associated BSs of users in IG, whereas the LSFD method considers only the BS connected to a single user within each UG. This confirms the effectiveness of our method. Although our method suffers a little performance loss compared with LLR-M, it still benefits from low computational cost, as it avoids access, channel acquisition and signal detection in advance.

Finally, we show the design and implementation complexities required by the proposed user-centric detection for UCN mMIMO system, and compare them with those in the optimal MMSE detection for network mMIMO system in Fig. 8 and Fig. 9, respectively. In particular, we let $I_{\mathrm{ig}} = 24$ and $\bar{B}_{\mathrm{cbs}} = 2$ for the user-centric detection, as the UCD under this configuration has shown satisfactory performance according to previous simulation results. From the figures, the design and implementation complexities of the UCD are much smaller than that of the optimal MMSE detector, while can achieve comparable performance. To be specific, the design complexity of the UCD is about $\frac{1}{89}$ of the optimal MMSE

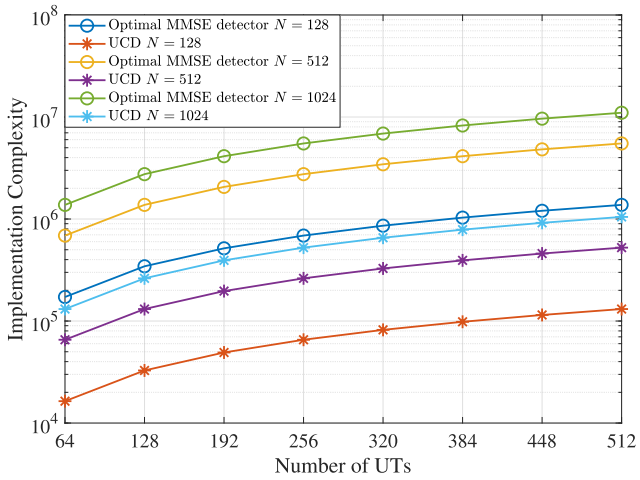


Fig. 9. Comparison of implementation complexities between the user-centric detector and the optimal MMSE detector.

detector when $N = 128$ and $U = 256$. Moreover, the design complexity of the UCD is only $\frac{1}{180}$ of the optimal MMSE detector when $N = 1024$ and $U = 512$. Additionally, the implementation complexity of the UCD is about $\frac{1}{10}$ of the optimal MMSE detector. In fact, as the number of antennas of each BS increases and the size of the system expands, the user-centric detection for the UCN mMIMO system is expected to exhibit even greater advantages.

V. Conclusion

In this paper, we have investigated the signal detection for UCN mMIMO systems. First, leveraging the inherent interference sparsity, we revealed that the detection performance of the MMSE detector can be guaranteed using the MF outputs of the intra-group users and a limited number of interfering users per UG. We showed that the detection performance of each UG is determined by a limited number of BSs with the BS connection sparsity. With both the interference and BS connection sparsities, we proposed the user-centric detection approach for the UCN mMIMO system. By first organizing the PSUs of each BS into distinct UGs, we detailed the process for constructing the IG and BC for each UG in UCN mMIMO systems. The user-centric detection operates in a two-stage process. For each UG, local MF is performed at each associated BS using local information in the first stage. In the second stage, group-wise IC is conducted at the PSBS with information exchanged from ASBSs. The simulation results verified the effectiveness and superiority of the proposed user-centric detection for UCN mMIMO systems.

Appendix A
Proof for Proposition 1

To demonstrate that the detector separately designed in each subset $\mathcal{U}_m$ remains optimal, it suffices to show that the MSE performance is consistent with (9). Obviously, we have

$$\mathrm{MSE}_{\mathrm{tot}} = \sum_{i \in \mathcal{U}} \mathrm{MSE}_i, \tag{47}$$

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which can be easily verified with (9) and (10). Therefore, we just need to prove that the MSE of user i is consistent with (10).

Consider the signal detection for user i in the user subset $\mathcal{U}_m$. For brevity, we denote the MMSE detector for user i that is designed within $\mathcal{U}_m$ as $\mathbf{v}_i^\star(\mathcal{U}_m)$. From (6), the MSE performance of user i under $\mathbf{v}_i^\star(\mathcal{U}_m)$ is given by

$$\begin{aligned}&\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{U}_m)\right)\\&=1+p_i\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{R}\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{h}_i-2p_i\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{h}_i\\&\overset{(a)}{=}1-p_i\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{h}_i+\sum_{j\notin\mathcal{U}_m}p_ip_j\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{h}_j\mathbf{h}_j^H\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{h}_i,\end{aligned}\tag{48}$$

where

$$\mathbf{R}=\sum_{j\in\mathcal{U}}p_j\mathbf{h}_j\mathbf{h}_j^H+\sigma_z^2\mathbf{I},\quad\mathbf{R}_{\mathcal{U}_m}=\sum_{j\in\mathcal{U}_m}p_j\mathbf{h}_j\mathbf{h}_j^H+\sigma_z^2\mathbf{I},\tag{49}$$

and (a) follows from $\mathbf{R}=\mathbf{R}_{\mathcal{U}_m}+\sum_{j\notin\mathcal{U}_m}p_j\mathbf{h}_j\mathbf{h}_j^H$. Let $\mathcal{U}_{m,\ell}=\mathcal{U}_m\cup\{\ell\}$, where $\ell\notin\mathcal{U}_m$. Then. the MSE performance under $\mathbf{v}_i^\star(\mathcal{U}_{m,\ell})$ is given by

$$\begin{aligned}&\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{U}_{m,\ell})\right)\\&=1+p_i\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_{m,\ell}}^{-1}\mathbf{R}\mathbf{R}_{\mathcal{U}_{m,\ell}}^{-1}\mathbf{h}_i-2p_i\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_{m,\ell}}^{-1}\mathbf{h}_i\\&=1-p_i\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_{m,\ell}}^{-1}\mathbf{h}_i+\sum_{k\notin\mathcal{U}_{m,\ell}}p_ip_k\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_{m,\ell}}^{-1}\mathbf{h}_k\mathbf{h}_k^H\mathbf{R}_{\mathcal{U}_{m,\ell}}^{-1}\mathbf{h}_i\\&\overset{(b)}{=}1-p_i\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{h}_i+p_ip_\ell\frac{\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{h}_\ell\mathbf{h}_\ell^H\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{h}_i}{1+\mathbf{h}_\ell^H\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{h}_\ell}\\&\quad+\sum_{k\notin\mathcal{U}_{m,\ell}}p_ip_k\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_{m,\ell}}^{-1}\mathbf{h}_k\mathbf{h}_k^H\mathbf{R}_{\mathcal{U}_{m,\ell}}^{-1}\mathbf{h}_i,\end{aligned}\tag{50}$$

where (b) follows from $(\mathbf{A}+\mathbf{b}\mathbf{b}^H)^{-1}=\mathbf{A}^{-1}-\frac{\mathbf{A}^{-1}\mathbf{b}\mathbf{b}^H\mathbf{A}^{-1}}{1+\mathbf{b}^H\mathbf{A}^{-1}\mathbf{b}}$.

We first prove the sufficiency, showing that if $\mathbf{h}_j^H\mathbf{h}_\ell=0,\ell\in\mathcal{U}_{m,\ell},\ell\notin\mathcal{U}_m$, then $\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{U}_{m,\ell})\right)=\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{U}_m)\right)$ holds. As $\mathbf{h}_j^H\mathbf{h}_\ell=0$, we have $\mathbf{R}_{\mathcal{U}_m}\mathbf{h}_\ell=\sum_{j\in\mathcal{U}_m}p_j\mathbf{h}_j\mathbf{h}_j^H\mathbf{h}_\ell+\sigma_z^2\mathbf{h}_\ell=\sigma_z^2\mathbf{h}_\ell$. Note that $\mathbf{R}_{\mathcal{U}_m}$ is positive definite, hence, the following equations

$$\begin{aligned}&\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{R}_{\mathcal{U}_m}\mathbf{h}_\ell=\mathbf{h}_\ell\\&\Rightarrow\sigma_z^2\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{h}_\ell=\mathbf{h}_\ell\Rightarrow\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{h}_\ell=\frac{1}{\sigma_z^2}\mathbf{h}_\ell\end{aligned}\tag{51}$$

always hold. Hence, we always have

$$\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{h}_\ell=\frac{1}{\sigma_z^2}\mathbf{h}_i^H\mathbf{h}_\ell=0,\ell\notin\mathcal{U}_m.\tag{52}$$

Similarly, we can prove that

$$\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_{m,\ell}}^{-1}\mathbf{h}_k=0,k\notin\mathcal{U}_{m,\ell}.\tag{53}$$

By substituting (52) and (53) into (48) and (50), we can reach

$$\begin{aligned}\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{U}_m)\right)&=\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{U}_{m,\ell})\right)\\&=1-p_i\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{h}_i,\end{aligned}\tag{54}$$

which is the MSE performance that only considers the signals from $\mathcal{U}_m\backslash\{i\}$ as the interference.

We now proceed to prove the necessity, that is, for $\forall\ell\notin\mathcal{U}_m$, if $\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{U}_{m,\ell})\right)=\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{U}_m)\right)$, $\mathbf{h}_j\mathbf{h}_\ell=0,j\in\mathcal{U}_m,\ell\notin\mathcal{U}_m$, always holds. Note that (48) and (50) always hold. By observing (48) and (50), we can see that the $\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{U}_{m,\ell})\right)=\mathrm{MSE}\left(\mathbf{v}_i^\star(\mathcal{U}_m)\right)$ holds for $\forall i\in\mathcal{U}_m$ when $\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_m}^{-1}\mathbf{h}_\ell=0$ and $\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_{m,\ell}}^{-1}\mathbf{h}_k=0,i\in\mathcal{U}_m,\ell\notin\mathcal{U}_m,k\notin\mathcal{U}_{m,\ell}$. Let $\mathbf{H}_{\mathcal{U}_m}$ and $\mathbf{\Lambda}_{\mathcal{U}_m}$ denote the stacked channel matrix and diagonal transmit power matrix of users in $\mathcal{U}_m$, respectively. Then, the condition can be equivalently rewritten as

$$\begin{aligned}&\mathbf{\Lambda}_{\mathcal{U}_m}^{\frac{1}{2}}\mathbf{H}_{\mathcal{U}_m}^H\left(\mathbf{H}_{\mathcal{U}_m}\mathbf{\Lambda}_{\mathcal{U}_m}\mathbf{H}_{\mathcal{U}_m}^H+\sigma_z^2\mathbf{I}\right)^{-1}\mathbf{h}_\ell=\mathbf{0}\\&\Rightarrow\left(\mathbf{\Lambda}_{\mathcal{U}_m}^{\frac{1}{2}}\mathbf{H}_{\mathcal{U}_m}^H\mathbf{H}_{\mathcal{U}_m}\mathbf{\Lambda}_{\mathcal{U}_m}^{\frac{1}{2}}+\sigma_z^2\mathbf{I}\right)^{-1}\mathbf{\Lambda}_{\mathcal{U}_m}^{\frac{1}{2}}\mathbf{H}_{\mathcal{U}_m}^H\mathbf{h}_\ell=\mathbf{0}\\&\Rightarrow\mathbf{H}_{\mathcal{U}_m}^H\mathbf{h}_\ell=\mathbf{0}.\end{aligned}\tag{55}$$

This completes the proof.

Appendix B
Proof for Proposition 2

We prove (18) by contradiction. Assuming

$$\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_1})\right)<\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_2})\right).\tag{56}$$

As the MMSE problem is strict convex, $\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_2})$ is the unique solution to minimize the MSE under $\mathcal{U}_{m_2}$, which implies

$$\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_2})\right)<\mathrm{MSE}\left(\bar{\mathbf{v}}_i(\mathcal{U}_{m_2})\right)\tag{57}$$

always holds for any other $\bar{\mathbf{v}}_i(\mathcal{U}_{m_2})$. Let

$$\bar{\mathbf{v}}_i^\#(\mathcal{U}_{m_2})=[\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_1});0],\tag{58}$$

we will have

$$\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\#(\mathcal{U}_{m_2})\right)<\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_2})\right)\tag{59}$$

according to the assumption (56), which contradicts (57). Therefore, the assumption should be rejected and this completes the proof for (18).

Next, we proceed to prove that $\mathbf{h}_\ell^H\mathbf{h}_j=0,\forall j\in\mathcal{U}$, is both sufficient and necessary for the equality in (18). The sufficiency is a direct consequence of Proposition 1. For the necessity, we need to prove that if $\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_1})\right)=\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_2})\right)$ holds, $\mathbf{h}_\ell^H\mathbf{h}_j=0,\forall j\in\mathcal{U}$, which can also be demonstrated by contradiction. Assuming $\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_1})\right)=\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_2})\right)$ holds when $\mathbf{h}_\ell^H\mathbf{h}_j\neq0$ for some $j\in\mathcal{U}$. As, $\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_1})\right)=\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_2})\right)$, we have

$$\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\#(\mathcal{U}_{m_2})\right)=\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_2})\right).\tag{60}$$

To be specific,

$$\begin{aligned}&\mathrm{MSE}\left(\bar{\mathbf{v}}_i^\star(\mathcal{U}_{m_2})\right)\\&=1-p_i\mathbf{h}_i^H\mathbf{H}_{\mathcal{U}_{m_2}}\bar{\mathbf{R}}_{\mathcal{U}_{m_2}}^{-1}\mathbf{H}_{\mathcal{U}_{m_2}}\mathbf{h}_i\\&=1-p_i\mathbf{h}_i^H[\mathbf{H}_{\mathcal{U}_{m_1}},\mathbf{h}_\ell]\begin{bmatrix}\mathbf{A}_{11}&\mathbf{A}_{12}\\\mathbf{A}_{21}&\mathbf{A}_{22}\end{bmatrix}^{-1}\begin{bmatrix}\mathbf{H}_{\mathcal{U}_{m_1}}^H\\\mathbf{h}_\ell^H\end{bmatrix}\mathbf{h}_i\\&=1-p_i\mathbf{h}_i^H[\mathbf{H}_{\mathcal{U}_{m_1}},\mathbf{h}_\ell]\begin{bmatrix}\mathbf{A}_{11}^{-1}+\mathbf{B}_{11}&\mathbf{B}_{12}\\\mathbf{B}_{21}&\mathbf{B}_{22}\end{bmatrix}\begin{bmatrix}\mathbf{H}_{\mathcal{U}_{m_1}}^H\\\mathbf{h}_\ell^H\end{bmatrix}\mathbf{h}_i\end{aligned}\tag{61}$$

where $\bar{\mathbf{R}}_{\mathcal{U}_{m_2}}=\bar{\mathbf{H}}_{\mathcal{U}_{m_2}}^H\mathbf{\Lambda}\bar{\mathbf{H}}_{\mathcal{U}_{m_2}}+\sigma_z^2\mathbf{H}_{\mathcal{U}_{m_2}}^H\mathbf{H}_{\mathcal{U}_{m_2}}$,$\mathbf{A}_{11}=\bar{\mathbf{R}}_{\mathcal{U}_{m_1}}$, $\mathbf{A}_{12}=\mathbf{A}_{21}^H=\bar{\mathbf{H}}_{\mathcal{U}_{m_1}}\mathbf{\Lambda}\mathbf{H}^H\mathbf{h}_\ell$,$\mathbf{A}_{22}=\mathbf{h}_\ell^H(\mathbf{H}\mathbf{\Lambda}\mathbf{H}^H+\sigma_z^2\mathbf{I})\mathbf{h}_\ell$, $\mathbf{B}_{11}=\mathbf{A}_{11}^{-1}\mathbf{A}_{12}\mathbf{A}_x^{-1}\mathbf{A}_{21}\mathbf{A}_{11}^{-1}$,$\mathbf{B}_{12}=-\mathbf{A}_{11}^{-1}\mathbf{A}_{12}\mathbf{A}_x^{-1}$, $\mathbf{B}_{21}=\mathbf{B}_{12}^H$ and $\mathbf{B}_{22}=\mathbf{A}_x^{-1}$ with $\mathbf{A}_x=\mathbf{A}_{22}-\mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}$. By

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observing the above equations, $\mathrm{MSE}\left(\bar{\mathbf{v}}_i^{\star}\left(\mathcal{U}_{m_2}\right)\right)$ can not be guaranteed to equal $\mathrm{MSE}\left(\bar{\mathbf{v}}_i^{\#}\left(\mathcal{U}_{m_2}\right)\right)$ if there exists $\mathbf{h}_\ell^H\mathbf{h}_j = 0, j \in \mathcal{U}$. Therefore, the assumption should be rejected and this completes the proof for the necessity.

Appendix C
Proof for Proposition 3

From (48) and (50), we have

$$\begin{aligned}
&\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{U}_{m_1}\right)\right) - \mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{U}_{m_2}\right)\right)\\
&= \sum_{j\notin\mathcal{U}_{m_1}} p_i p_j \mathbf{h}_i^H \mathbf{R}_{\mathcal{U}_{m_1}}^{-1}\mathbf{h}_j\mathbf{h}_j^H\mathbf{R}_{\mathcal{U}_{m_1}}^{-1}\mathbf{h}_i - p_i p_\ell \frac{\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_{m_1}}^{-1}\mathbf{h}_\ell\mathbf{h}_\ell^H\mathbf{R}_{\mathcal{U}_{m_1}}^{-1}\mathbf{h}_i}{1+\mathbf{h}_\ell^H\mathbf{R}_{\mathcal{U}_{m_1}}^{-1}\mathbf{h}_\ell}\\
&\quad - \sum_{j\notin\mathcal{U}_{m_2}} p_i p_j \mathbf{h}_i^H \mathbf{R}_{\mathcal{U}_{m_2}}^{-1}\mathbf{h}_j\mathbf{h}_j^H\mathbf{R}_{\mathcal{U}_{m_2}}^{-1}\mathbf{h}_i\\
&\le \frac{p_i p_\ell \alpha_{\ell,\ell}}{1+\alpha_{\ell,\ell}}|\alpha_{i,\ell}|^2 + \sum_{j\notin\mathcal{U}_{m_2}} \frac{p_i p_j |\alpha_{i,\ell}\alpha_{j,\ell}|^2}{|1+\alpha_{\ell,\ell}|^2}\\
&\quad + \sum_{j\notin\mathcal{U}_{m_2}} 2\Re\left\{\frac{p_i p_j \alpha_{i,\ell}\alpha_{\ell,j}\alpha_{j,i}}{1+\alpha_{\ell,\ell}}\right\} \qquad (62)
\end{aligned}$$

where $\alpha_{m,n} \triangleq \mathbf{h}_m^H\mathbf{R}_{\mathcal{U}_{m_1}}^{-1}\mathbf{h}_n$. Note that

$$\begin{aligned}
&|\alpha_{i,\ell}| = |\mathbf{h}_i^H\mathbf{R}_{\mathcal{U}_{m_1}}^{-1}\mathbf{h}_\ell|\\
&= \frac{1}{\sigma_z^2}|\mathbf{h}_i^H\left(\mathbf{I} - \mathbf{H}_{\mathcal{U}_{m_1}}\left(\sigma_z^2\mathbf{I} + \mathbf{H}_{\mathcal{U}_{m_1}}^H\mathbf{H}_{\mathcal{U}_{m_1}}\right)^{-1}\mathbf{H}_{\mathcal{U}_{m_1}}^H\right)\mathbf{h}_\ell|\\
&\le \frac{1}{\sigma_z^4}\left(|\mathbf{h}_i^H\mathbf{h}_\ell| + |\mathbf{h}_i^H\mathbf{H}_{\mathcal{U}_{m_1}}\left(\sigma_z^2\mathbf{I} + \mathbf{H}_{\mathcal{U}_{m_1}}^H\mathbf{H}_{\mathcal{U}_{m_1}}\right)^{-1}||\mathbf{H}_{\mathcal{U}_{m_1}}^H\mathbf{h}_\ell|\right)\\
&\le c_2\delta, \qquad (63)
\end{aligned}$$

where $c_2 = \frac{1}{\sigma_z^2}(1+U_1|\mathbf{h}_i^H\mathbf{H}_{\mathcal{U}_{m_1}}\left(\sigma_z^2\mathbf{I} + \mathbf{H}_{\mathcal{U}_{m_1}}^H\mathbf{H}_{\mathcal{U}_{m_1}}\right)^{-1}|) > 0$ is a known scalar and the equality holds if $\mathbf{H}_{\mathcal{U}_{m_1}}^H\mathbf{h}_\ell = \mathbf{0}$. Bysubstituting (63) to (62), we can obtain (25). Additionally, as $\delta \to 0$, $\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{U}_{m_2}\right)\right) \to \mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{U}_{m_1}\right)\right)$. This completes the proof.

Appendix D
Proof for Proposition 4

First, we focus on the proof of (32), which can be established by contradiction. Assuming

$$\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_1}\right)\right) < \mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right) \qquad (64)$$

holds. As the MMSE problem is strict convex, $\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)$ is the only solution that can minimize the MSE when considering $\mathcal{I}_m$ as the IG and $\mathcal{B}_{m_2}$ as the BC. Therefore,

$$\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right) < \mathrm{MSE}\left(\mathbf{v}_i\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right) \qquad (65)$$

holds for any $\mathbf{v}_i\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right) \neq \mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right) \in \mathbb{C}^{NB_{m_2}} \times 1$, where $B_{m_2} = \mathrm{card}\left(\mathcal{B}_{m_2}\right)$. Let

$$\mathbf{v}_i^{\#}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right) = \left[\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_1}\right);\mathbf{0}\right], \qquad (66)$$

we will have

$$\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right) > \mathrm{MSE}\left(\mathbf{v}_i^{\#}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right) \qquad (67)$$

according to (64), which contradicts with (65). Therefore, the assumption should be rejected and we always have $\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_1}\right)\right) \ge \mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right)$. This completes the proof of (32).

Next, we prove that the $\mathbf{H}_{\mathcal{I}_m,\ell} = \mathbf{0}$ is both sufficient and necessary for the equality in (32). For the sufficiency, as $\mathbf{H}_{\mathcal{I}_m,\ell} = \mathbf{0}$, we have $\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_2}} = [\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_1}};\mathbf{0}]$ and $\mathbf{h}_{i,\mathcal{B}_{m_2}} = [\mathbf{h}_{i,\mathcal{B}_{m_1}};\mathbf{0}]$, where $\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_2}}$ is the stacked channel matrix of all users in $\mathcal{I}_m$ to all the BSs in $\mathcal{B}_{m_2}$ and $\mathbf{h}_{i,\mathcal{B}_{m_2}}$ is the channel vector from user i to all the BSs in $\mathcal{B}_{m_2}$. Let $\mathbf{\Lambda}_{\mathcal{I}_m}$ denote the diagonalized transmit power of all users in$\mathcal{I}_m$. Then, we have

$$\begin{aligned}
&\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right)\\
&= 1 - p_i\mathbf{h}_{i,\mathcal{B}_{m_2}}^H\left(\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_2}}\mathbf{\Lambda}_{\mathcal{I}_m}\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_2}}^H + \sigma_z^2\mathbf{I}_{B_{m_2}N}\right)^{-1}\mathbf{h}_{i,\mathcal{B}_{m_2}}\\
&= 1 - p_i[\mathbf{h}_{i,\mathcal{B}_{m_1}}^H,\mathbf{0}]\begin{bmatrix}\mathbf{X} & \mathbf{0}\\ \mathbf{0} & \frac{1}{\sigma_z^2}\mathbf{I}\end{bmatrix}\begin{bmatrix}\mathbf{h}_{i,\mathcal{B}_{m_1}}\\ \mathbf{0}\end{bmatrix}\\
&= 1 - p_i\mathbf{h}_{i,\mathcal{B}_{m_1}}^H\mathbf{X}\mathbf{h}_{i,\mathcal{B}_{m_1}} = \mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_1}\right)\right). \qquad (68)
\end{aligned}$$

where $\mathbf{X} = \left(\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_1}}\mathbf{\Lambda}_{\mathcal{I}_m}\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_1}}^H + \sigma_z^2\mathbf{I}\right)^{-1}$. This completes the proof for the sufficiency.

For the necessity, we need to prove that when $\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right) = \mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_1}\right)\right)$ holds, $\mathbf{H}_{\mathcal{I}_m,\ell}$ should be $\mathbf{0}$. This can also be demonstrated by contradiction. Assuming $\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right) = \mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_1}\right)\right)$ holds when $\mathbf{H}_{\mathcal{I}_m,\ell} \neq \mathbf{0}$. Then, we have

$$\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right) = \mathrm{MSE}\left(\mathbf{v}_i^{\#}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right), \qquad (69)$$

where $\mathbf{v}_i^{\#}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)$ has been defined in (66). Recall that $\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)$ is the only solution to the MMSE problem we consider. Therefore, (69) indicates that $\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right) = \mathbf{v}_i^{\#}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)$ must hold for arbitrary $\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_2}}$. Note that

$$\begin{aligned}
&\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\\
&= \sqrt{p_i}\left(\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_2}}\mathbf{\Lambda}_{\mathcal{I}_m}\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_2}}^H + \sigma_z^2\mathbf{I}\right)^{-1}\mathbf{h}_{i,\mathcal{B}_{m_2}}. \qquad (70)
\end{aligned}$$

Obviously, it cannot be guaranteed that $\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right) = \mathbf{v}_i^{\#}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)$ when $\mathbf{H}_{\mathcal{I}_m,\ell} \neq \mathbf{0}$. This contradicts with (69). Therefore, the assumption should be rejected and $\mathbf{H}_{\mathcal{I}_m,\ell}$ must be $\mathbf{0}$. This completes the proof for the necessity.

Finally, we quantify the performance loss by removing BS ℓ from the BC of user i. Note that $\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_2}} = [\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_1}};\mathbf{H}_{\mathcal{I}_m,\ell}]$ and $\mathbf{h}_{i,\mathcal{B}_{m_2}} = [\mathbf{h}_{i,\mathcal{B}_{m_1}};\mathbf{h}_{i,\ell}]$. Let $\mathbf{A} = \mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_2}}\mathbf{\Lambda}_{\mathcal{I}_m}\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_2}}^H + \sigma_z^2\mathbf{I}_{B_{m_2}N}$. Note that

$$\begin{aligned}
\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right) &= 1 - p_i\mathbf{h}_{i,\mathcal{B}_{m_2}}^H\mathbf{A}^{-1}\mathbf{h}_{i,\mathcal{B}_{m_2}}\\
&= 1 - p_i\mathbf{h}_{i,\mathcal{B}_{m_2}}^H\begin{bmatrix}\mathbf{A}_{11} & \mathbf{A}_{12}\\ \mathbf{A}_{21} & \mathbf{A}_{22}\end{bmatrix}^{-1}\mathbf{h}_{i,\mathcal{B}_{m_2}} \qquad (71)
\end{aligned}$$

where $\mathbf{A}_{11} = \mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_1}}\mathbf{\Lambda}_{\mathcal{I}_m}\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_1}}^H + \sigma_z^2\mathbf{I}$, $\mathbf{A}_{12} = \mathbf{A}_{21}^H = \mathbf{H}_{\mathcal{I}_m,\mathcal{B}_{m_1}}\mathbf{\Lambda}_{\mathcal{I}_m}\mathbf{H}_{\mathcal{I}_m,\ell}^H$, $\mathbf{A}_{22} = \mathbf{H}_{\mathcal{I}_m,\ell}\mathbf{\Lambda}_{\mathcal{I}_m}\mathbf{H}_{\mathcal{I}_m,\ell}^H + \sigma_z^2\mathbf{I}$. More specifically, we have

$$\mathbf{A}^{-1} = \begin{bmatrix}\mathbf{A}_{11}^{-1} & \mathbf{0}\\ \mathbf{0} & \mathbf{0}\end{bmatrix} + \begin{bmatrix}\mathbf{B}_{11} & \mathbf{B}_{12}\\ \mathbf{B}_{21} & \mathbf{B}_{22}\end{bmatrix}, \qquad (72)$$

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where $\mathbf{B}_{11}=\mathbf{A}_{11}^{-1}\mathbf{A}_{12}\mathbf{A}_{x}^{-1}\mathbf{A}_{21}\mathbf{A}_{11}^{-1}$, $\mathbf{B}_{12}=-\mathbf{A}_{11}^{-1}\mathbf{A}_{12}\mathbf{A}_{x}^{-1}$, $\mathbf{B}_{21}=\mathbf{B}_{12}^{H}$ and $\mathbf{B}_{22}=\mathbf{A}_{x}^{-1}$ with $\mathbf{A}_{x}=\mathbf{A}_{22}-\mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}$. By substituting (72) into (71), we have

$$\begin{aligned}&\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_1}\right)\right)-\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right)\\&=p_i\mathbf{h}_{i,\mathcal{B}_{m_2}}^{H}\mathbf{B}\mathbf{h}_{i,\mathcal{B}_{m_2}}\\&=p_i(\mathbf{h}_{i,\mathcal{B}_{m_1}}^{H}\mathbf{B}_{11}\mathbf{h}_{i,\mathcal{B}_{m_1}}+2\Re\{\mathbf{h}_{i,\ell}^{H}\mathbf{B}_{21}\mathbf{h}_{i,\mathcal{B}_{m_1}}\}+\mathbf{h}_{i,\ell}^{H}\mathbf{B}_{22}\mathbf{h}_{i,\ell})\\&=\zeta_{m,\ell},\end{aligned}\tag{73}$$

where $\zeta_{m,\ell}\geq 0$ according to (32). This complete the proof of (33).

Appendix E
Proof for Proposition 5

As shown in (73),

$$\begin{aligned}&\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_1}\right)\right)-\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right)\\&=p_i(\mathbf{h}_{i,\mathcal{B}_{m_1}}^{H}\mathbf{B}_{11}\mathbf{h}_{i,\mathcal{B}_{m_1}}+2\Re\{\mathbf{h}_{i,\ell}^{H}\mathbf{B}_{21}\mathbf{h}_{i,\mathcal{B}_{m_1}}\}+\mathbf{h}_{i,\ell}^{H}\mathbf{B}_{22}\mathbf{h}_{i,\ell})\end{aligned}\tag{74}$$

For $\mathbf{h}_{i,\mathcal{B}_{m_1}}^{H}\mathbf{B}_{11}\mathbf{h}_{i,\mathcal{B}_{m_1}}$ in (73), we have

$$\begin{aligned}\left\|\mathbf{h}_{i,\mathcal{B}_{m_1}}^{H}\mathbf{B}_{11}\mathbf{h}_{i,\mathcal{B}_{m_1}}\right\|&\leq\left\|\mathbf{h}_{i,\mathcal{B}_{m_1}}^{H}\right\|\left\|\mathbf{B}_{11}\mathbf{h}_{i,\mathcal{B}_{m_1}}\right\|\\&\leq\lambda_{\max}\left(\mathbf{B}_{11}\right)\left\|\mathbf{h}_{i,\mathcal{B}_{m_1}}\right\|^2.\end{aligned}\tag{75}$$

With the relationship between $\mathbf{B}_{11}$ and $\mathbf{H}_{\ell}$, we can obtain that $\lambda_{\max}\left(\mathbf{B}_{11}\right)\leq c_1\phi$, where $c_1>0$ is a scalar. By repeatedly applying such inequalities to (73), finally, we have

$$\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right)\geq\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_1}\right)\right)-p\phi,\tag{76}$$

where $p\geq 0$. As $\phi\to 0$, $\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_2}\right)\right)\to\mathrm{MSE}\left(\mathbf{v}_i^{\star}\left(\mathcal{I}_m,\mathcal{B}_{m_1}\right)\right)$. This completes the proof.

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